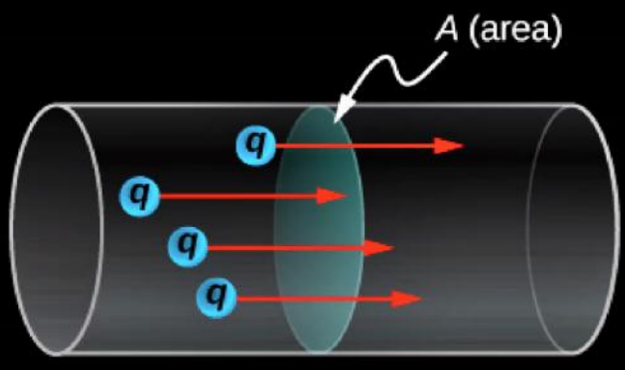
ELECTRIC CURRENT:

Rate of flow of charge through any area is called current. If ΔQ amount of charge crosses a surface in time Δt, we define average current as

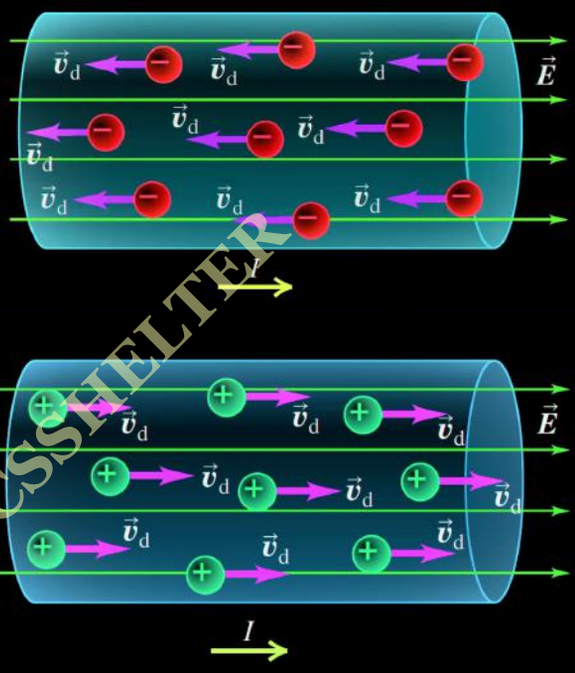
$$I_{avg} = \Delta Q/\Delta t$$

and instantaneous current is defined as:

$$I = \lim_{\Delta t \to 0} (\Delta Q/\Delta t) = dQ/dt.$$

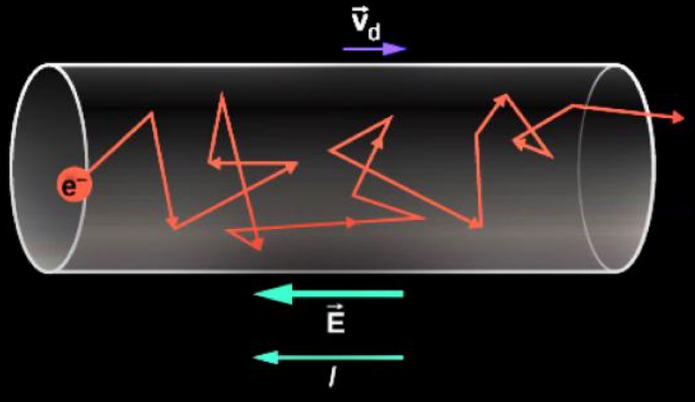


In metals, current develops due to presence of electric field created with help of battery. Electrons move against this electric field and conventional current is considered to be in a direction opposite to flow of electrons. If an electron is moving leftward, it is equivalent to a positron moving rightward for purpose of calculating current.



MODEL OF CONDUCTION IN METALS:

In absence of applied electric field, electrons behave like gas particles in random thermal motion, such that their kinetic energy is given as $KE = 3/2\ kT$ where T is temperature. The rms speed of this electron gas is of the order $10^6\ ms^{-1}$.



The electrons are in constant random motion and thus on an average total charge crossing any section of a conductor is zero. However when a field is created in the conductor, it accelerates the electrons (though often some time electrons "collide" with lattice points and amongst themselves and the velocities get randomized again). The overall effect of applied electric field is that electrons on an average begin to advance on an average by a velocity of order of few mm/sec to few cm/sec. This velocity is known as drift velocity.

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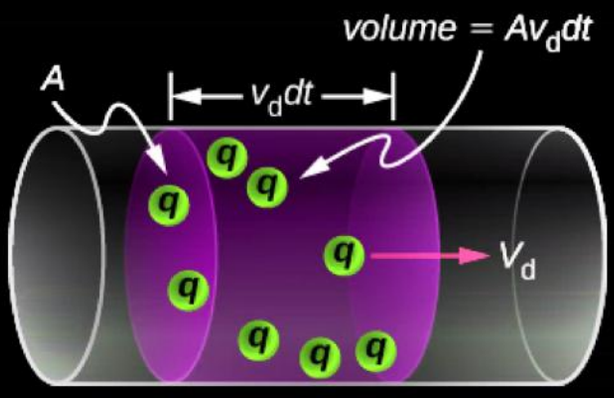
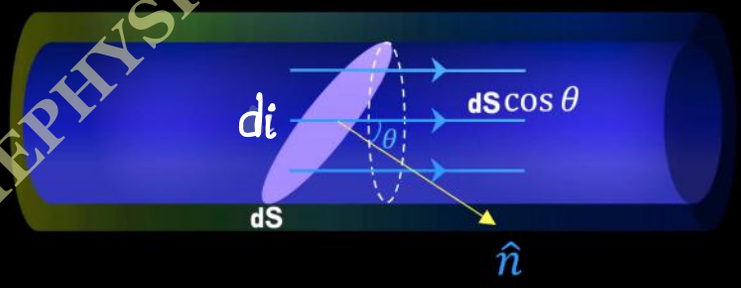
Relation Between Current And Drift Velocity: ($\vec{A} \parallel \vec{v}_d$)



Let n be the number of free charge carriers in a conductor per unit volume. Now consider a cross section of the conductor $\perp$ to drift velocity, in a time dt, the charge carriers advance by a distance $v_d dt$, thus the number of charge carriers crossing a cross section $A \perp$ to v_d is $A v_d dt \times n$. Now if e is the charge of each charge carrier, the charge crossed in time dt is given as:

$$dq = n e A v_d dt \Rightarrow dq/dt = n e A v_d$$

Thus $\boxed{i = n e A v_d}$



Extending the above analysis to elemental areas in arbitrary orientations, we can write elemental current through an elemental patch $\vec{dS}$ as

$$di = n e \vec{v}_d \cdot \vec{dS}$$

The quantity $n e \vec{v}_d$ is known as current density $\vec{J}$, thus we can write current through elemental patch as

$$\boxed{di = \vec{J} \cdot \vec{dS}}$$

Current Density Vector: Amount of charge flowing per unit time per unit area held perpendicular to local drift velocity gives the magnitude of current density vector, and the direction of current density vector is same as that of the local drift velocity. Now if an area dS is held $\perp$ to drift velocity, the elemental current through it is given as $di = n e v_d dS$, thus $J = di/dS$ i.e $J = n e v_d$ or

$$\boxed{\vec{J} = n e \vec{v}_d}$$

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Mean Relaxation Time(τ): Recall that apart from a directed drift, the charge carriers in a conductor are also undergoing random collisions. After every random collision, the velocity of charge carriers also gets randomised whose average value is zero. The mean time elapsed since last collision for the charge carriers in a conductor is called mean relaxation time. Here is a statement from Purcell and Morin's book about mean relaxation time that might appear surprising, confusing and even shocking:

8 You may think the average time between collisions would have to be equal to the *sum* of the *average time since the last collision* and the *average time to the next.* That would be true if collisions occurred at absolutely regular intervals, but they don't. They are independent random events, and for such the above statement, paradoxical as it may seem at first, is true. Think about it. The question does not affect our main conclusion, but if you unravel it you will have grown in statistical wisdom; see Exercise 4.23. (*Hint:* If one collision doesn't affect the probability of having another – that's what *independent* means – it can't matter whether you start the clock at some arbitrary time, or at the time of a collision.)

Here the author is referring to the fact that mean relaxation time is also the average time elapsed between two successive collisions which is also equal to mean time elapsed since last collision and also equal to mean time remaining for next collision.

→ This statement really pained me and kept me disturbed for quite a few days. However if you think carefully you can understand this in terms of rolling a die. Suppose you are keeping a track of the event of rolling a 6. The mean number of rolls required to get the next 6 is independent of the number that you just rolled. Thus mean time required for rolling next 6 is same irrespective of the fact that whether you just rolled a 6 or any other random number. Thus mean rolls between 6 and next 6 is same as mean number of rolls from an arbitrary time to next 6. Mean relaxation time is governed by thermal velocity of electrons (recall that thermal velocity is of the order of $10^6 ms^{-1}$, while drift velocity is in mm/sec. With increase in temperature, mean relaxation time decreases due to frequent collisions.

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Relation Between Electric Field And Drift Velocity:

Consider all free electrons in a conductor. Let $\vec{u}_i$ be the velocity of i^{th} electron just after its previous collision and let t_i be the time elapsed since its last collision, the by kinematics of uniformly accelerated motion:

$$\vec{v}_i = \vec{u}_i + \vec{a}t_i = \vec{u}_i + \frac{\vec{E}e}{m}t_i$$

Thus $\vec{v}_d = \langle \vec{v}_i \rangle = \langle \vec{u}_i \rangle + \frac{\vec{E}e}{m}\langle t_i \rangle$

But since velocity gets randomised after every collision, for a large sample $\langle \vec{u}_i \rangle = 0$, thus:

$$\boxed{\vec{v}_d = \frac{\vec{E}e}{m}\tau}$$

→ H.C. Verma Sir's book gives $v_d = \frac{Ee}{2m}\tau$ which is not correct (it is coming from a historical error committed Paul Drude who proposed a model for conduction in metals but was off by a factor of 2 in his analysis. We would get a factor of 2 in denominator if the electron collisions were not independent).

→ With increase in temperature drift velocity decreases due to reduction in mean relaxation time.

Resistance: The ratio of potential difference and current through a conductor is called resistance. If the current is directly proportional to Voltage applied, we call such conductors as ohmic conductors. Mathematically, $\boxed{R = V/I}$ here V is voltage applied across conductor and I is current through the conductor. In the next section, we shall show that metals are ohmic conductors at room temperature. We can create non-ohmic conductors by doping the semiconductors about which we will study in great detail in a later chapter.

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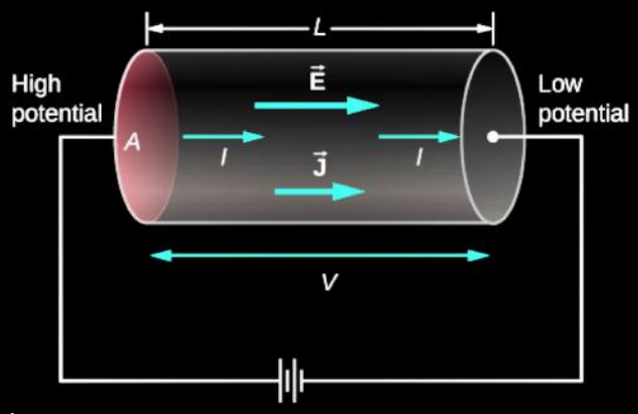
OHM'S LAW: Consider a metallic conductor carrying a current as shown, Now, $V = EL$ and $I = neAv_d = neA \times \frac{Ee}{m}\tau$



$\Rightarrow I = \frac{ne^2\tau}{m}\frac{AV}{L}$ or

$\frac{V}{I} = \frac{m}{ne^2\tau}\frac{L}{A}$, now in this expression it can be easily seen that RHS is a constant for a conductor of a given dimensions and material. Here:

m = mass of electrons

n = number density of free electrons (constant for a given metal)

e = electronic charge

τ = mean relaxation time (constant at a given temperature for a given metal)

L = length of conductor

A = Area of cross section of conductor.

RESISTIVITY (ρ) AND CONDUCTIVITY (σ): The factor $m/ne^2\tau$ that we obtained during derivation of ohm's law is a material property known as resistivity i.e.

$$\rho = m/ne^2\tau$$

Its reciprocal is called conductivity i.e $\sigma = 1/\rho = ne^2\tau/m$

Using the expression for resistivity we can write:

$\frac{V}{I} = \rho\frac{L}{A} = R$ Thus resistance of any prismatic conductor is given as $R = \rho\frac{L}{A}$, Here L is along the current and A is $\perp$ to current.

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Relation Between Electric Field, Current Density And Resistivity:

Recall the expression we obtained in our derivation of ohm's law:

$I = neAv_d : neA \times \frac{Ee}{m}\tau \Rightarrow J = I/A = E \times \frac{ne^2\tau}{m}$,

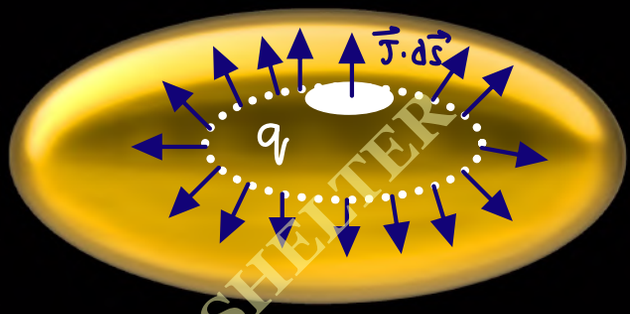
but $\rho = m/ne^2\tau$, thus $J = E/\rho$, also $\boxed{\vec{J} = \vec{E}/\rho = \sigma\vec{E}}$

Continuity Equation:

Recall for any patch $di = \vec{J}\cdot d\vec{S}$, thus if we have a completely closed surface, we can write:

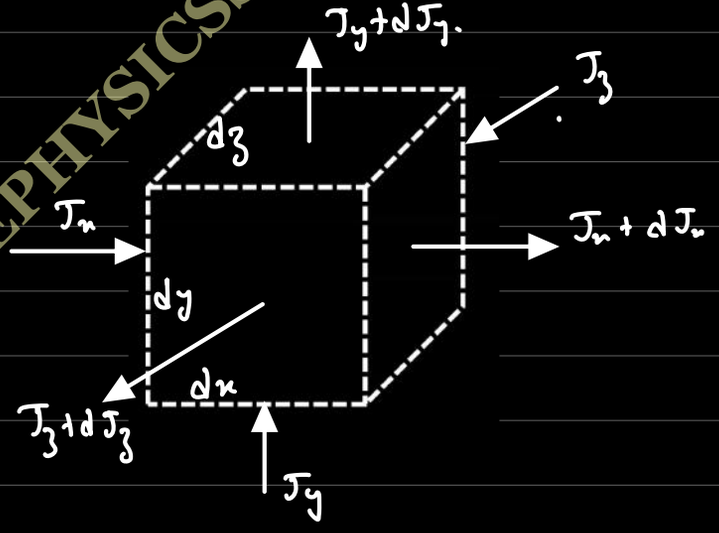
$$\oint \vec{J}\cdot d\vec{S} = i = -\frac{d}{dt}(q_{in})$$

thus $\boxed{\oint \vec{J}\cdot d\vec{S} = -\frac{dq_{in}}{dt}}$



Here we have used charge conservation i.e charge flowing out must be equal to rate of depletion inside the boundary.

→ For steady state conduction, $\oint \vec{J}\cdot d\vec{S} = 0$ (∵ There is no net accumulation or depletion).



→ We can also apply continuity equation for an elemental volume shown above thus:

$$(J_x + dJ_x)\,dy\,dz - J_x\,dy\,dz + (J_y + dJ_y)(dx\,dz) - J_y\,dx\,dz + (J_z + dJ_z)\,dx\,dy - J_z\,dx\,dy = -\frac{d}{dt}(\rho\,dx\,dy\,dz)$$

Rearranging we get: $\frac{\partial J_x}{\partial x} + \frac{\partial J_y}{\partial y} + \frac{\partial J_z}{\partial z} = -\frac{d\rho}{dt}$

or $\boxed{\vec{\nabla}\cdot\vec{J} = -\frac{d\rho}{dt}}$

→ For steady state conduction, $\vec{\nabla}\cdot\vec{J} = 0$

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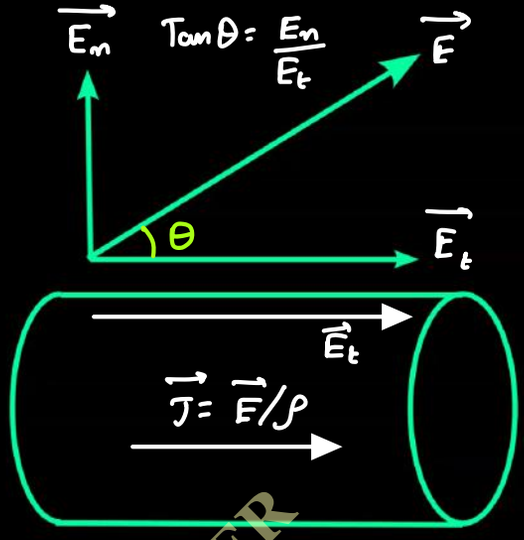
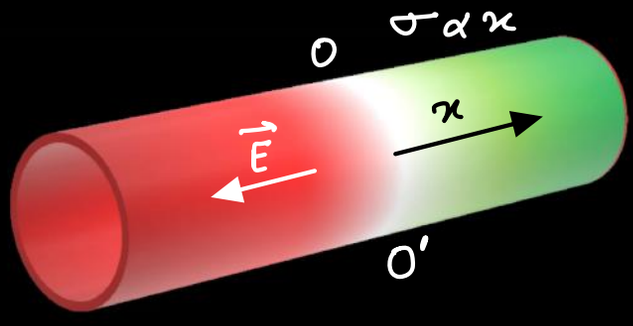
ELECTRIC FIELD IN CURRENT CARRYING CONDUCTORS:

In the last section we showed that for steady state conduction $\oint \vec{J}\cdot d\vec{s} = 0$, now for a homogeneous conductor this means: $\oint \sigma \vec{E}\cdot d\vec{s} = 0$ or $\sigma \oint \vec{E}\cdot d\vec{s} = 0$, thus for a conductor carrying a steady current, net charge enclosed in volume must be zero, then the question arises how does the field arise in a conductor. It turns out that when a circuit is switched on, there is a very brief period of time during which charge accumulation occurs on a conductor surface. This charge accumulation on surface creates a field in the conductor which causes the current to flow along the conductor. The tangential field just outside the conductor is exactly equal to tangential field just inside the conductor, besides the tangential field, the conductor will in general also have a normal component of electric field which may be towards or away from the surface depending on the local charge density being positive or negative.



→ Recall that we have already seen one such distribution which creates a uniform field inside the conductor despite the total net charge being zero (We had solved a related problem during our discussions on Gauss' law).



GENERALIZED OHM'S LAW: Recall that we had shown that in presence of an electric field $\vec{E}$, $\vec{J} = \vec{E}/\rho = \sigma\vec{E}$, Here $\vec{E}$ was the electrostatic force per unit charge pushing the charge through the conductor, however the pushing force need not be of electrostatic nature alone it could be of some other nature as well (for instance we discussed how in a chemical battery, chemical forces are also responsible for pushing the charges). Let $\vec{E}^*$ be the

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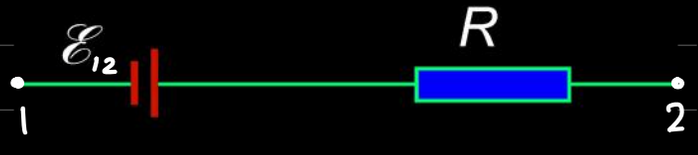
force per unit charge of non electrostatic origin then our generalized ohm's law at any point in a conductor becomes

$$\boxed{\vec{J} = (\vec{E} + \vec{E}^*)/\rho = \sigma(\vec{E} + \vec{E}^*)}$$

Integral Form Of Ohm's Law For A Subcircuit:

Rearranging the equation obtained in previous section:



$\rho \vec{J} = (\vec{E} + \vec{E}^*)$, Now suppose we advance a displacement $d\vec{l}$ along a current stream tube then taking dot product with $d\vec{l}$ on both sides:

$$\rho \vec{J} \cdot d\vec{l} = (\vec{E} + \vec{E}^*) \cdot d\vec{l} = \rho \frac{i}{A} dl = i \rho \frac{dl}{A} = i\,dR \quad (1)$$

Here dR is the resistance of element of length dl.

Now integrating on both sides we get:

$$\int_1^2 \vec{E} \cdot d\vec{l} + \int_1^2 \vec{E}^* \cdot d\vec{l} = \int_1^2 i\,dR$$

Now recall that $-\int \vec{E} \cdot d\vec{l} = V_2 - V_1$ (Potential difference)

and $\int \vec{E}^* \cdot d\vec{l}$ is EMF between two points, also in case of discrete resistances, integration can be replaced by summation thus:

$$\boxed{V_1 - V_2 + \mathcal{E}_{12} = \Sigma IR}$$ ← Ohm's law for any branch, here

I is considered as positive from 1 to 2, $\mathcal{E}$ is positive if its driving polarity is from 1 to 2.

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Ohm's Law For Complete Loops (Kirchoff's Loop Law):

In a previous section we got:

$$(\vec{E} + \vec{E}^*) \cdot d\vec{l} = i\,dR$$

Now integrating over closed paths we get:

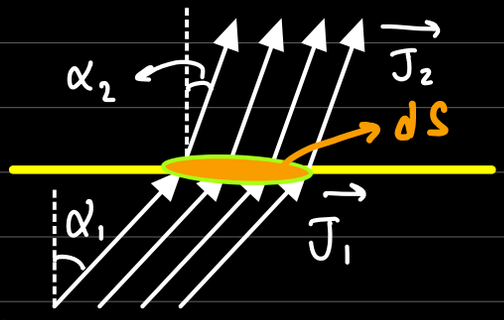
$$\oint \vec{E} \cdot d\vec{l} + \oint \vec{E}^* \cdot d\vec{l} = \oint i\,dR$$

But Since $\vec{E}$ is conservative, $\therefore \oint \vec{E} \cdot d\vec{l} = 0$, thus,

$$\oint \vec{E}^* \cdot d\vec{l} = \oint i\,dR \quad \text{or} \quad \Sigma \mathcal{E} = \Sigma iR$$ for discrete batteries and resistor circuits. $\mathcal{E}$ is considered as positive if driving polarity is along the the loop traversal, and i is also considered positive if along the direction of loop traversal.

Example: Irodov: 3.164:

Demonstrate that the law of refraction of direct current lines at the boundary between two conducting media has the form $\tan\alpha_2 / \tan\alpha_1 = \sigma_2/\sigma_1$, where σ_1 and σ_2 are the conductivities of the media, α_2 and α_1 are the angles between the current lines and the normal of the boundary surface.



Solution: In case of steady state conduction, there is no accumulation of charge on any patch, thus inflow must equal outflow thus:

$$\vec{J}_1 \cdot d\vec{S} = \vec{J}_2 \cdot d\vec{S} \Rightarrow J_1 \cos\alpha_1 = J_2 \cos\alpha_2 \quad (1)$$

Now tangential field is continuous across boundary and $\vec{J} = \sigma\vec{E}$ thus

$E_t = J_t/\sigma$, thus $J_1 \sin\alpha_1/\sigma_1 = J_2 \sin\alpha_2/\sigma_2$ (2)

Now (2)/(1) gives:

$$\frac{\tan\alpha_1}{\sigma_1} = \frac{\tan\alpha_2}{\sigma_2}$$

or $$\frac{\tan\alpha_2}{\tan\alpha_1} = \frac{\sigma_2}{\sigma_1}$$ QED.

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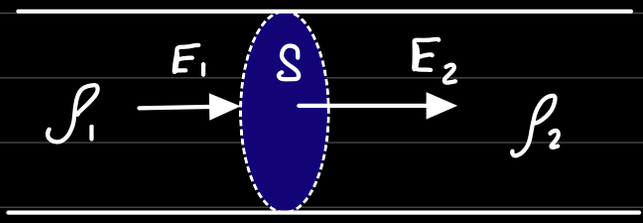
Example : Irodov 3.165 :

Two cylindrical conductors with equal cross-sections and different resistivities ρ_1 and ρ_2 are put end to end. Find the charge at the boundary of the conductors if a current I flows from conductor 1 to conductor 2.



Solution: Since current on left side and right side of interface is same, $\therefore$:

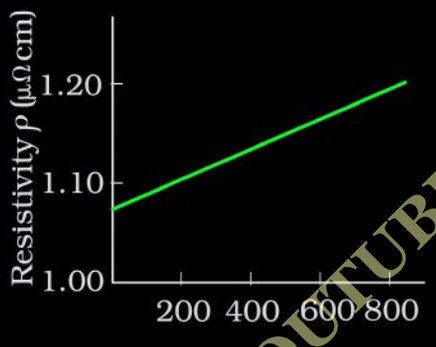
$$\frac{E_1}{\rho_1} = \frac{E_2}{\rho_2} = \frac{I}{S}$$

Now imagine a pill shaped Gaussian surface around the interface, applying Gauss law to this Gaussian surface,

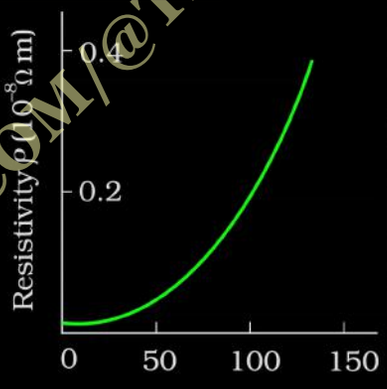
$$E_2 S - E_1 S = \frac{q_{in}}{\epsilon_0} = (\rho_2 - \rho_1) I \quad (\because E_1 S = \rho_1 I)$$

$$\Rightarrow \quad q_{in} = \epsilon_0 (\rho_2 - \rho_1) I \text{ Ans}$$

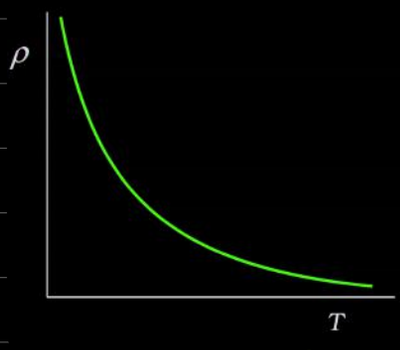
Effect Of Temperature On Resistivity :



NICHROME



COPPER



SEMICONDUCTOR

Recall that $\rho = m/ne^2\tau$, in metals n does not change with temperature however τ reduces with temperature, accordingly near room temperature, we can approximately write

$$\boxed{\rho = \rho_0 (1 + \alpha \Delta T)}$$ Here ΔT is rise in temperature.

For most metals coefficients of thermal expansion are about two orders magnitude smaller than temperature coefficient of resistance thus for resistances also we can write $R = R_0 (1 + \alpha \Delta T)$

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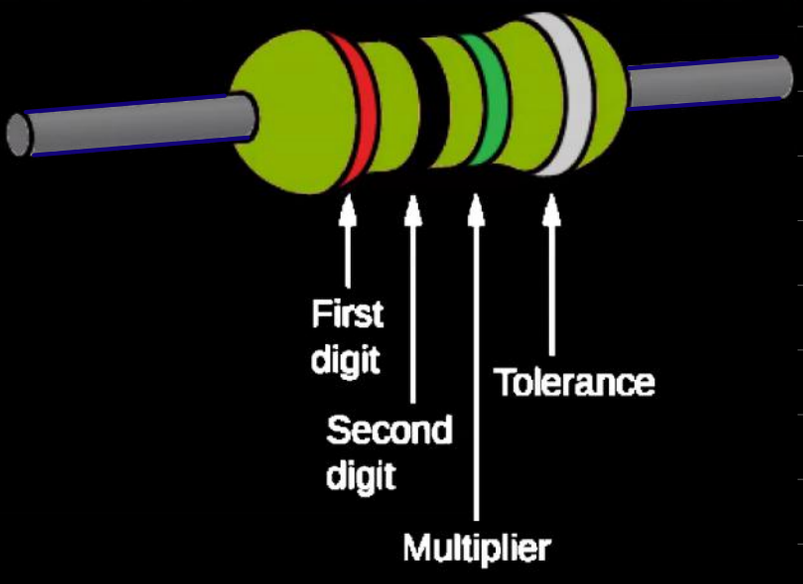
here α is called temperature coefficient of resistance. In case of semiconductors, n itself varies with temperature (increases with increase in temperature) and this effect is predominant and thus resistance of semiconductors decreases with rise in temperature.

Colour Coding Of Carbon Resistors:



$R = 20 \times 10^5\ \Omega \pm 10\%$
$R = 2.0M\ \Omega \pm 0.2M\ \Omega$

Color	Digit	Multiplier
Black	0	10^0
Brown	1	10^1
Red	2	10^2
Orange	3	10^3
Yellow	4	10^4
Green	5	10^5
Blue	6	10^6
Violet	7	10^7
Gray	8	10^8
White	9	10^9

Tolerance	
Gold	5%
Silver	10%

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For use on small sized printed circuit boards (PCBs) very small sized carbon resistors are available whose resistance can be inferred from the nature of ring pattern on them. Typically there are four rings on such resistors, which can be read as explained by the graphic above. (Sometimes fourth ring is missing which corresponds to a tolerance of $\pm 20\%$). Following Mnemonic can be used for memorizing the ring Scheme:

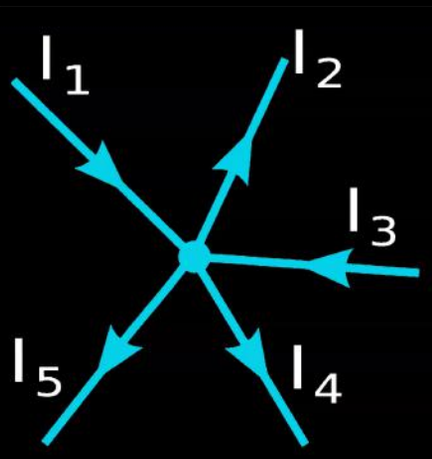
B B ROY GREAT BRITAIN VERY GOOD WIFE GOOD SON
Tolerance

B → Black, Brown, Blue
G → Green, Grey, Gold

Sometimes a fifth ring might be present which represents temperature coefficient.

Kirchoff's Rules For Resistor Circuits:

Kirchoff's Junction Law:



In absence of accumulation, net current entering a junction is equal to current leaving the junction. Or alternatively algebraic sum of all currents entering a junction is zero, thus:

According to first statement:

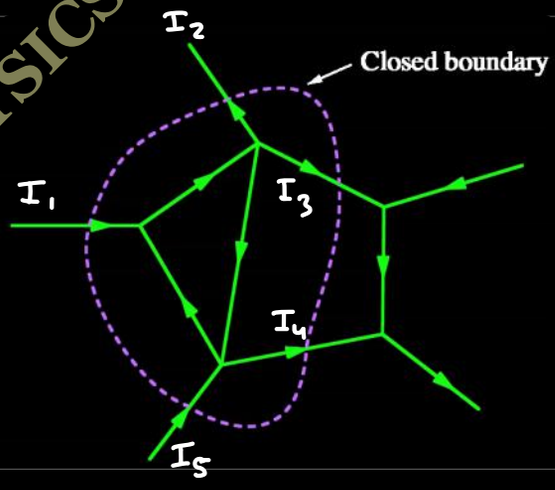
$$I_1 + I_3 = I_2 + I_4 + I_5$$ and

according to second statement $$I_1 - I_2 + I_3 - I_4 - I_5 = 0$$

Kirchoff's junction law simply based on conservation of charge.

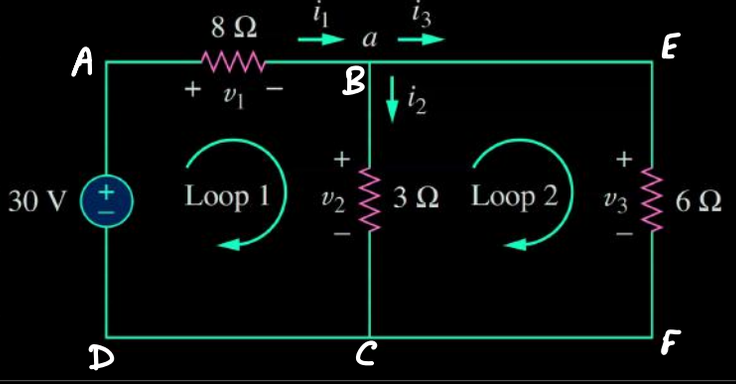
→ Not just at junctions, the charge conservation principle can be applied to every boundary in which no charge accumulation is taking place. Thus for adjacent figure,

$$I_1 + I_5 = I_2 + I_3 + I_4.$$



Kirchoff's Loop Law:

Algebraic sum of all potential drops in a circuit is zero. For purpose of applying Kirchoff's loop law if we cross a resistor while moving along the current, we write a $+iR$ for the potential drop and we write a $-iR$ if we traval against the current. For batteries, the convention is same as we dissussed in the lectures on capacitor circuits. Thus for loop ABCDA:



$$8i_1 + 3i_2 - 30 = 0$$ Similarly for loop 2: $$6i_3 - 3i_2 = 0$$

(Also by junction law, $i_1 = i_2 + i_3$).

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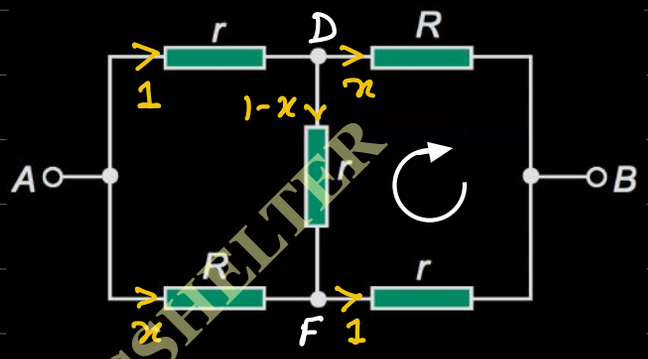
Effective Resistance Of A Network: If a network of resistors is connected across a battery of EMF V and it draws a current i from it then we define effective resistance as simply $R_{eff} = V/i$. Thus for finding effective resistance, we can connect a network across a 1 Volt battery and find the reciprocal of current drawn.

Example: Irodov 3.187:

Find the effective resistance of the circuit between points A and B.



Solution: Resistance of circuit network is absolute and does not depend on the Voltage or current. The currents get scaled up in proportion to the voltage of the battery. Let for some voltage V of the battery currents flow as shown in orange notation. Note that the circuit has a symmetry about a point and we have used it to reduce the number of unknowns.

Now applying loop law to DBFD:

$$Rx - xr - (1-x)r = 0 \implies x = 2r/R+r.$$

Now total potential drop between A and B (which is also battery EMF):

$$V = 1 \times r + x R \quad (\because \text{travelling along ADB})$$

$$\implies V = \frac{r(r+3R)}{r+R}$$

Now total current drawn from battery, $i = 1+x = \frac{3r+R}{r+R}$

Now effective resistance $R_{eff} = V/i = \frac{r(r+3R)}{(3r+R)}$ Ans.

Node Voltage Method: Just as we did in capacitance circuits, here also we can take voltage variables at various nodes and satisfy junction law at various nodes to find the node potentials. Since current flows depend on potential differences rather than

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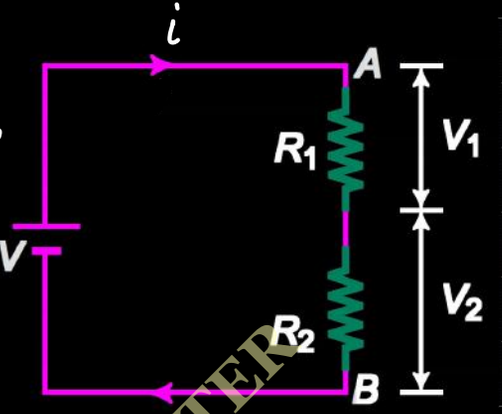
absoluted potential, here also we can choose potential at one of the nodes as zero and consider it as reference node relative to which potentials of all other nodes are defined.

RESISTORS IN SERIES: Two or more resistors are said to be connected in series if they carry same current by virtue of configuration (i.e they are on same branch without any distributaries in between). In series it can be readily seen that:



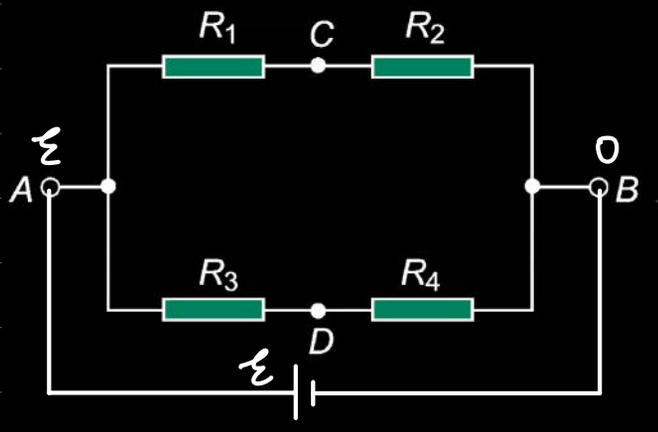
$$R_{eff} = R_1 + R_2$$

Also since potential drop across any resistor is iR, thus in series the potential drop is in direct ratio of the resistances thus

$$V_1 = \frac{R_1}{R_1 + R_2} V$$

For adjacent circuit.

EXAMPLE: For the circuit shown in adjacent figure, assuming potential of node B as zero, find potentials of nodes C and D and hence find relation between resistances for this potential difference to be zero



Solution: R_1 and R_2 are in series with a total potential drop of $\mathcal{E}$. Thus: $V_C = \left(\frac{R_2}{R_1 + R_2}\right)\mathcal{E}$ Ans

Similarly, $V_D = \frac{R_4}{R_4 + R_3}\mathcal{E}$ Ans.

Now for making nodes equipotential: $V_C = V_D$

$$\Rightarrow \quad \frac{R_4}{R_4 + R_3} = \frac{R_2}{R_2 + R_1}$$

Now Doing $\frac{N}{D-N}$ on both sides

$R_4/R_3 = R_2/R_1$ Ans.

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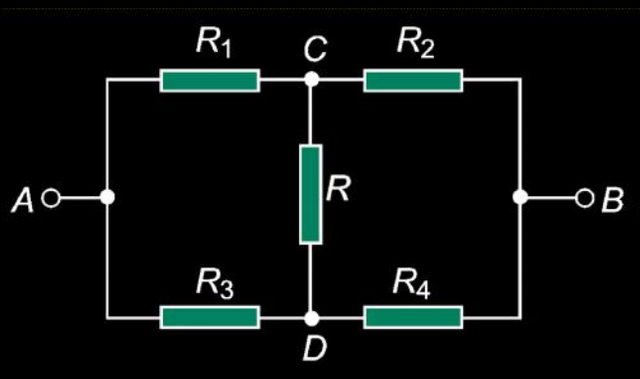
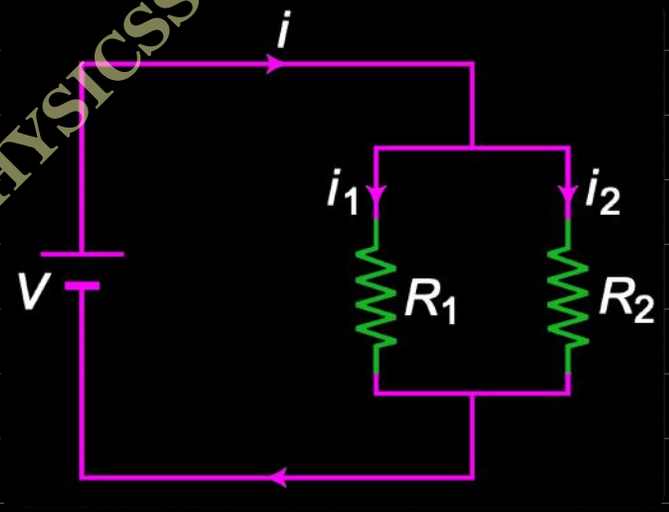
Balanced Wheatstone Bridge:



Consider the circuit shown in adjacent figure. In the previous section we saw that in such a configuration if $R_1/R_2 = R_3/R_4$ then the points C and D become equipotential and any resistor (or anything else except a battery) connected between them draws no charge or current hence, the branch CD becomes redundant under such circumstances. When this happens, we say that the circuit is a balanced wheatstone bridge.

Resistors In Parallel:



Two or more resistors are said to be connected in parallel if by virtue of configuration, they have same potential drops (i.e they are directly connected between same nodes).

Here we can say that:

$$i = i_1 + i_2 = \frac{V}{R_1} + \frac{V}{R_2}$$

$$\Rightarrow \frac{i}{V} = \frac{1}{R_{eff}} = \frac{1}{R_1} + \frac{1}{R_2}, \text{ thus in parallel: } \boxed{\frac{1}{R_{eff}} = \frac{1}{R_1} + \frac{1}{R_2}}$$

Now since $i = V/R$, in parallel combination, the total current supplied to the node is distributed in inverse ratio of resistances. Thus:

$$\boxed{i_1 = \frac{1/R_1}{1/R_1 + 1/R_2} \times i}$$

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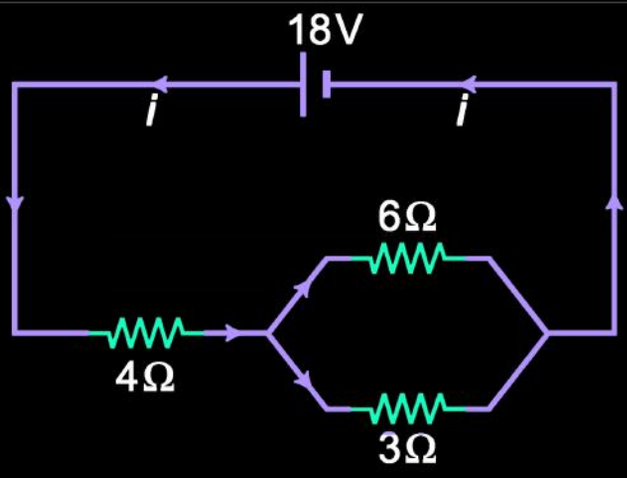
Example: For the circuit shown in adjacent figure, find :

(a) Effective resistance

(b) Currents and Voltages for each resistor.



Solution: $R_{eff} = (6||3) \text{ Ser } 4 = 2 \text{ Ser } 4$

$\Rightarrow R_{eff} = 6\,\Omega$ Ans.

Now total current drawn $i = 18/6 = 3$ amp.

Thus $i_4 = 3A$ Ans.

Now among $6\,\Omega$ and $3\,\Omega$ this current distributes in inverse ratio of resistances, thus $i_6 = \dfrac{1/6}{1/6+1/3} \times 3 = 1A$ Amp.

Now $i_2 = i - i_6 = 2A$ Ans.

Now $V_4 = i_4 \times 4 = 12$ Volts Ans

$V_3 = i_3 \times 3 = 6$ Volts $= V_6$ Ans.

Circuit Theorems:

Linearity Property: In any circuit containing batteries and resistors, current or potential drop across any branch is just a linear combination of various batteries. Thus schematically if i and v be the current and potential drop across some branch then

$$i = f(R_i)\,\mathcal{E}_1 + g(R_i)\,\mathcal{E}_2 + h(R_i)\,\mathcal{E}_3 + \cdots$$

Similarly

$$v = \alpha(R_i)\,\mathcal{E}_1 + \beta(R_i)\,\mathcal{E}_2 + \gamma(R_i)\,\mathcal{E}_3 + \cdots$$

This property follows from the fact that when Kirchoff's loop law is applied to any loop in a circuit containing batteries and resistors, each equation is a linear equation.

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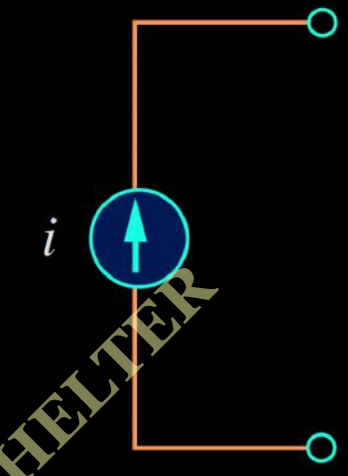
(This linearity is also followed in circuits containing only capacitors and ideal batteries, however not in a circuit simultaneous containing resistors and capacitors).

IDEAL CURRENT SOURCE: An ideal current source is the one which supplies a constant current irrespective of the network connected across its terminals. It can be imagined to be a battery of infinite EMF and infinite internal resistance such that the ratio of EMF and the internal resistance is a finite number Equal to current from the source. (Imagine a Battery of 4Billion volts and internal resistance 1 Billion Ω, for all relatively small external resistance, it will supply a constant current of 4A irrespective of resistance.



→ Potential drop across a current source can adjust according to the load network but current remains constant. (While for ideal battery, current can adjust but potential drop is constant)

LINEARITY WITH CURRENT SOURCES: Just as with voltage sources, the linearity property holds with current sources also (linearity also holds with mix of current and voltage sources so long as other elements are only resistors). Thus schematically:

$$i = f(R_i)I_1 + g(R_i)I_2 + h(R_i)I_3 + \cdots$$

Similarly

$$v = \alpha(R_i)I_1 + \beta(R_i)I_2 + \gamma(R_i)I_3 + \cdots$$

Here $I_1, I_2, I_3 \ldots$ etc are current sources currents and R_i is the i^{th} resistance in the network.

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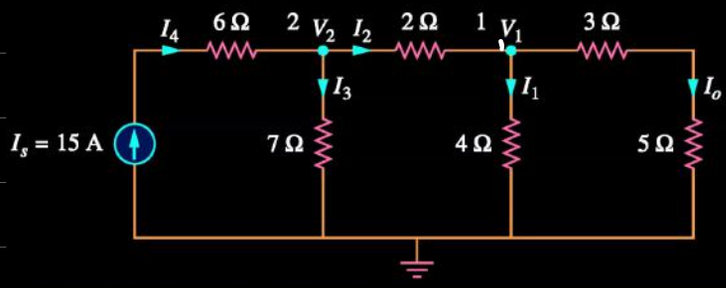
EXAMPLE:

Assume $I_o = 1A$ and use linearity to find the actual value of I_o in the shown adjacent figure.



Solution: Assuming $I_o = 1A$ we will find source current required to produce it and scale up I_o according to actual I_s. Now assuming $I_o = 1A$, $V_1 = 1(3+5) = 8V$

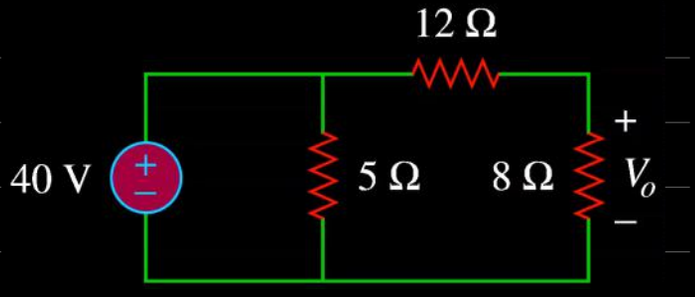
$$\Rightarrow I_1 = 8/4 = 2A \Rightarrow I_2 = I_1 + I_o = 3A \Rightarrow V_2 = V_1 + 2I_2$$

$$\Rightarrow V_2 = 14V \Rightarrow I_3 = V_2/7\Omega = 2A$$

$$\Rightarrow I_4 = I_2 + I_3 = 5A = I_s' \quad (\text{for } I_o = 1A)$$

However actual $I_s = 15A$ which is $3I_s'$ thus I_o must also be scaled by a factor of 3. Thus $I_o = 3A$ Ans.

Think Yourself: Assuming $V_o = 1V$ find the actual value of V_o using linearity.



Solution: If $V_o = 1V$ then drop across 12Ω will be $(12/8) \times 1 = 1.5V \Rightarrow$ Source voltage will have to be $1 + 1.5 = 2.5$ Volts, but actual source voltage is 40 Volt which is 16×2.5 thus actual V_o must also be 16V.

SUPERPOSITION THEOREM (BATTERIES): In any circuit containing only batteries and resistors the current is superposition of currents due to each battery acting alone (with all other batteries short circuited).
If batteries have internal resistances, they are treated just like external resistances for applying Superposition.

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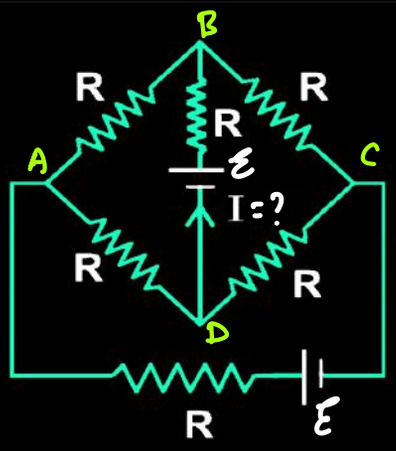
EXAMPLE: For the adjacent circuit find the current I using superposition theorem.

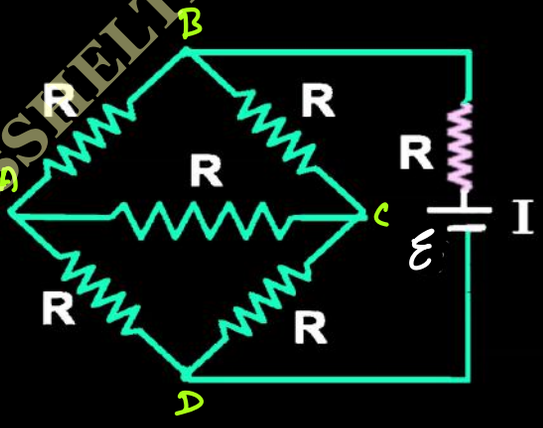
Hint: Upon considering one battery at a time try to spot wheatstone bridges.



Solution: First we ignore the battery in the vertical branch and we can clearly see the standard wheatstone configuration, thus due to lower branch battery nothing flows in branch BD.

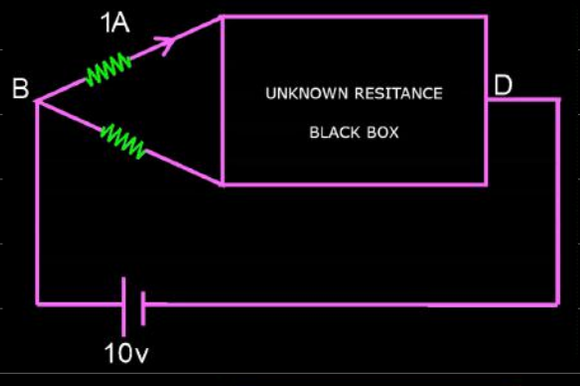
Now ignoring the battery in lower branch, the circuit can be redrawn as shown in adjacent figure, Now branch AC is redundant once again due to wheatstone bridge configuration. Now the circuit simply reduces to a series-parallel combination with $R_{eff} = 2R$

Thus $I = \mathcal{E}/2R$ Ans.



THINK YOURSELF: What would be the current if battery in vertical branch made 5 times and that in horizontal branch 100 times? Ans $5\mathcal{E}/2R$.

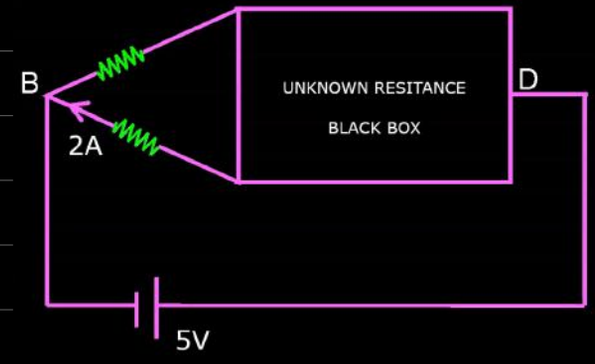
EXAMPLE: Consider a network of resistances being driven by a battery of 10 Volt in first instance and 5 Volts in second instance with known currents as shown. Find effective resistance between B and D.



Solution: Notice that in lower figure battery polarity is reversed. Thus with 10 V battery lower branch should be carrying 4A in rightward direction.

Thus net current drawn from 10V battery

$i = 4+1 = 5A \Rightarrow R = V/i = 10/5 = 2\Omega$ Ans.



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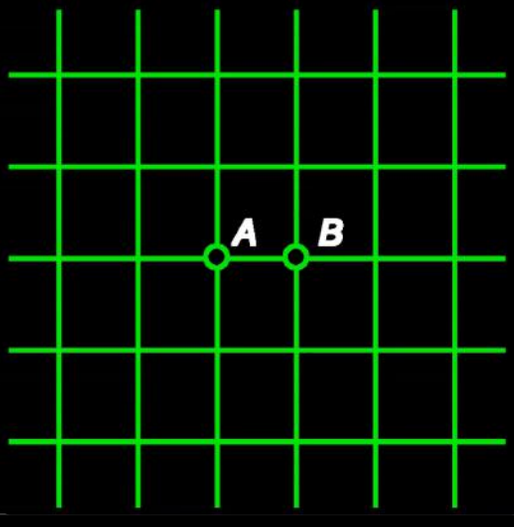
Superposition Theorem (Current Sources): If i_1 is current in a particular branch of a resistor circuit due to a current source I_1 acting alone and if i_2 is current in the same branch due to current source I_2 acting alone then if the current source I_1 is scaled by factor α and current source I_2 is scaled by a factor of β and both are simultaneously deployed then net current in that branch will be

$$i = \alpha i_1 + \beta i_2.$$

(Or simply stated, net current in any branch is superposition of currents due to each source individually. While considering effect of any current source, all other sources are replaced by open circuits).

Example: IRODOV 3.153

There is an infinite wire grid with square cells (See Fig.). The resistance of each wire between neighbouring joint connections is equal to R_0. Find the resistance R of the whole grid between points A and B.
Instruction. Make use of principles of symmetry and superposition.



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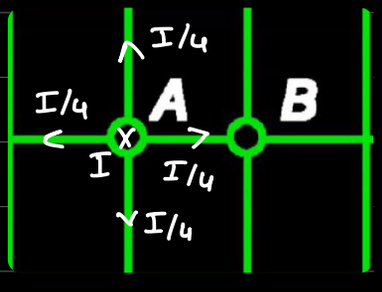
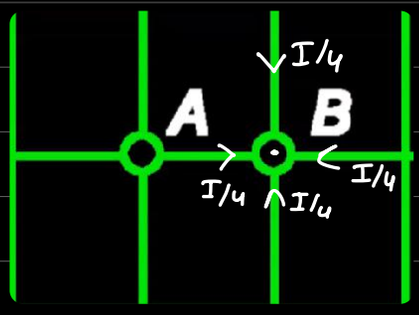
Solution: We shall model the current configuration as superposition of following operations:

1. Tapping in a current I from A while tapping out from infinity
2. Tapping in current from from infinity and tapping out from B.



Now when I current is tapped in from A, $I/4$ flows from A to B due to symmetry.

Similary when I current is tapped out from B (tapping in from ∞) once again a current I/4 flows from A to B.



Now superposition of two effects is that a current I is tapped in from A and tapped out from B with a current I/2 flowing in the direct resistor from A to B.

Thus potential difference across network: $V = I/2 \times R_0$

Now $R_{eff} = V/I = R_0/2$ ohms.

EXAMPLE: IRODOV 3.155:

A metal ball of radius a is surrounded by a thin concentric metal shell of radius b. The space between these electrodes is filled up with a poorly conducting homogeneous medium of resistivity ρ. Find the resistance of the inter electrode gap. Analyse the obtained solution at $b \to \infty$.

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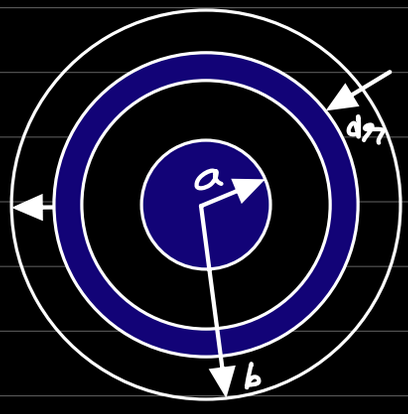
Solution: We can consider the system to be made of several concentric shells connected in series.



Thus, $dR = \rho \frac{dr}{4\pi r^2} \Rightarrow R = \int dR$

$$\Rightarrow R = \rho\left[\frac{1}{a} - \frac{1}{b}\right] = \frac{\rho(b-a)}{4\pi ab} \text{ ohms.}$$

Now when $b \to \infty$, $R = \rho/4\pi a$ ohms.

→ Recall that capacitance of the system is $C = 4\pi\epsilon_0 ab/(b-a)$. An interesting result is observed multiplying RC, we see that, $RC = \rho\epsilon_0$ (If there were dielectric $RC = \rho\epsilon$). This is not only true for spherical geometries but also for cylindrical and other geometries filled with homogeneous dielectric and having homogeneous resistivities. (We can model capacitance as series combination of infinite number of infinitesimal capacitors with conductors arranged along equipotentials).

Example: Irodov 3.157:

Two metal balls of the same radius a are located in a homogeneous poorly conducting medium with resistivity ρ. Find the resistance of the medium between the balls provided that the separation between them in much greater than the radius of the ball.

Solution: Once again we will model the situation by superposition.

We shall first tap in a current I from one ball and tap out at infinity and then tap in I from infinity and take out from the other ball.

Now tapping I from first ball,

the potential difference between balls ① and ② $V_1 = IR = I \times \frac{\rho}{4\pi a}$

(Ball 1 is at higher potential and ball ② at zero potential (being at infinite distance).

Now similarly when current I is tapped out from ball ②, its potential, $V_2 = -I\rho/4\pi a$, due to superposition effect:

potential difference between the balls, $V = V_1 - V_2 = \rho I/2\pi a$

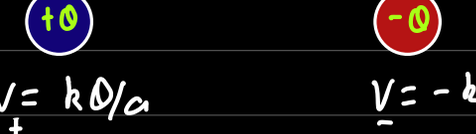
$\Rightarrow R_{eff} = \frac{V}{I} = \rho/2\pi a$ Ans.

→ Alternatively we could also use $RC = \rho\varepsilon_0$

Potential difference between spheres, $V = 2kQ/a$

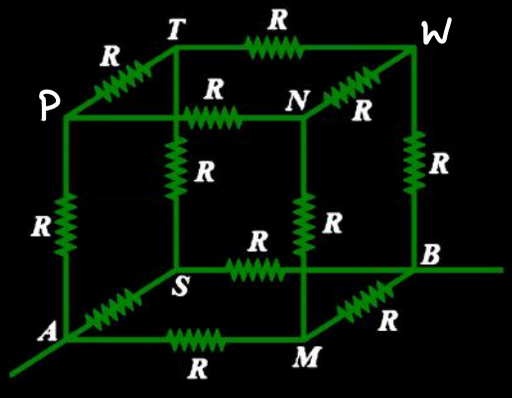
$\Rightarrow C = Q/V = 2\pi\varepsilon_0 a$

Now $RC = \rho\varepsilon_0 \Rightarrow R = \rho\varepsilon_0/C$



$\Rightarrow R = \rho\varepsilon_0/(2\pi\varepsilon_0 a) = \rho/2\pi a$ (This is same answer as above)

Example: Finding effective resistance of the shown cube network across face diagonal AB. (Use superposition method).



Solution: The situation is modelled using superpositions listed:

① Tap in I from A and tap out $I/8$

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from all vertices (including A also)

② Tap out I from B and tap in I/8 from all vertices (including B vertex).

Now due to first step, $I_{AM} = (I - I/8) \times \frac{1}{3} = 7/24\, I$, this

Now Current available for distribution between MN & MB $= \frac{7}{24} I - \frac{I}{8}$ i.e I/6, thus MB recieves I/12 (by symmetry).

Thus potential drop between A & B during the first step, :

$$V_1 = \frac{7}{24} IR + \frac{1}{12} IR = \frac{9}{24} IR$$

Similarly by symmetry we can say that same drop occurs in the second step, thus, $V_2 = \frac{9}{24} IR$, combining both steps,

net potential drop, $V = V_1 + V_2 = \frac{9}{12} IR = \frac{3}{4} IR \Rightarrow R_{eff} = V/I$

thus $R_{eff} = \frac{3}{4} R$ Ans.

(Recall for capacitor circuit we got $C_{eff} = \frac{4}{3} C$, this type of reciprocal relation can also be observed for results across body diagonal ($5R/6$) and edge ($7R/12$).

→ Think at home: Using this method figure out resistance across body diagonal.

→ We could also have solved this problem using symmetry (collapsing equipotential points to a single point and converting to series parallel circuit just as we did for capacitor cube).

EXAMPLE: 2025 points are arranged in space and each point is connected to every other point through a resistor R. Find effective resistance between any two points.

Solution: Input I current through a vertex (say vertex A) and extract a current I/2025 from each vertex

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(including A itself), in this process current flowing from A to another vertex (say B), $I_{AB} = \left(I - \frac{I}{2025}\right) \times \frac{1}{2024} = \frac{I}{2025}$

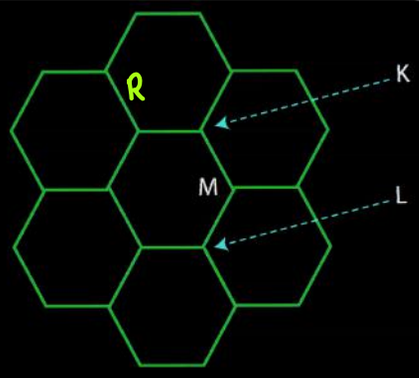
thus potential difference created in first step $V_1 = IR/2025$, similarly in second step same PD is created, thus total P.D.:

$V = V_1 + V_2 = 2IR/2025 \Rightarrow R_{eff} = V/I = 2R/2025$ Ans.

(Recall for capacitor network we had $C_{eff} = \frac{2025\,C}{2}$, once again we see that reciprocal relation holds good).

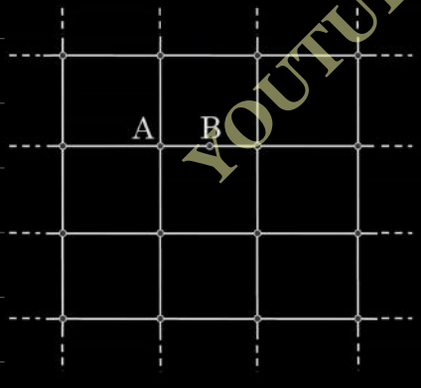
Think at home:

1. Find effective resistance across edge of a dodecahedron (Each edge has a resistance R) Ans (19/30)R

2. 

R_{eff} between K and L = ?
(Each edge is R, grid is infinite)
(Ans R)

3. In a cube each edge, body diagonal and face diagonal carries a resistor R. Find effective resistance across the edge. (Ans R/4)

4. 

Find effective resistance between A and B (each edge has a resistance R).
Ans: 3R/8.

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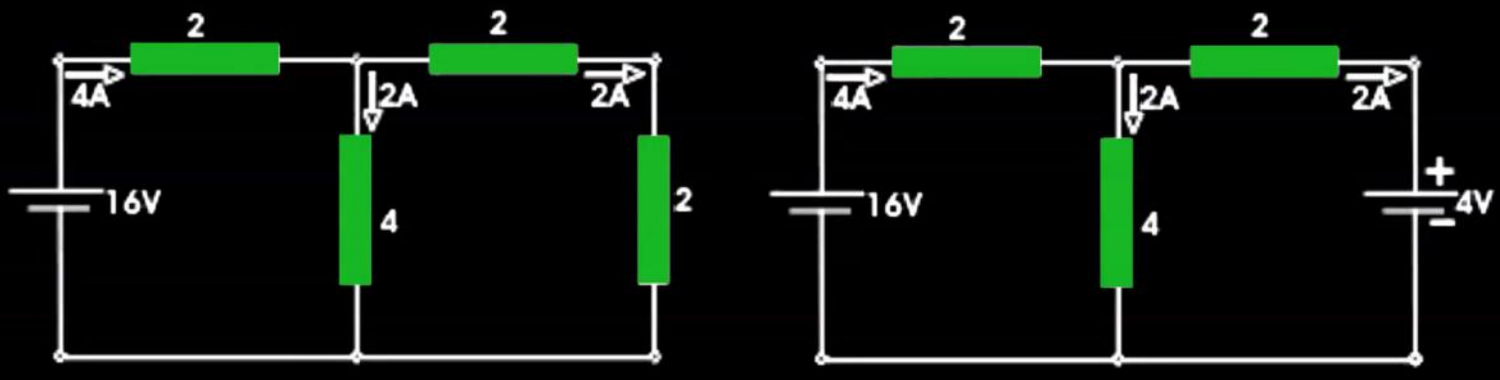
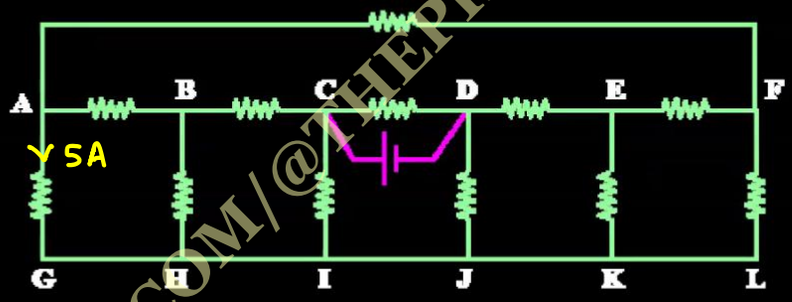
COMPENSATION THEOREM: Any Resistance in any network can be replaced by a voltage source that has zero internal resistance and a voltage equal to potential drop across the resistor being replaced due to the current flowing through it.



For instance in the circuit shown above, the 2Ω resistor has been replaced by a 4V battery in the second figure and one can readily verify that it makes no difference to the currents in the circuit (and hence also potential drops).

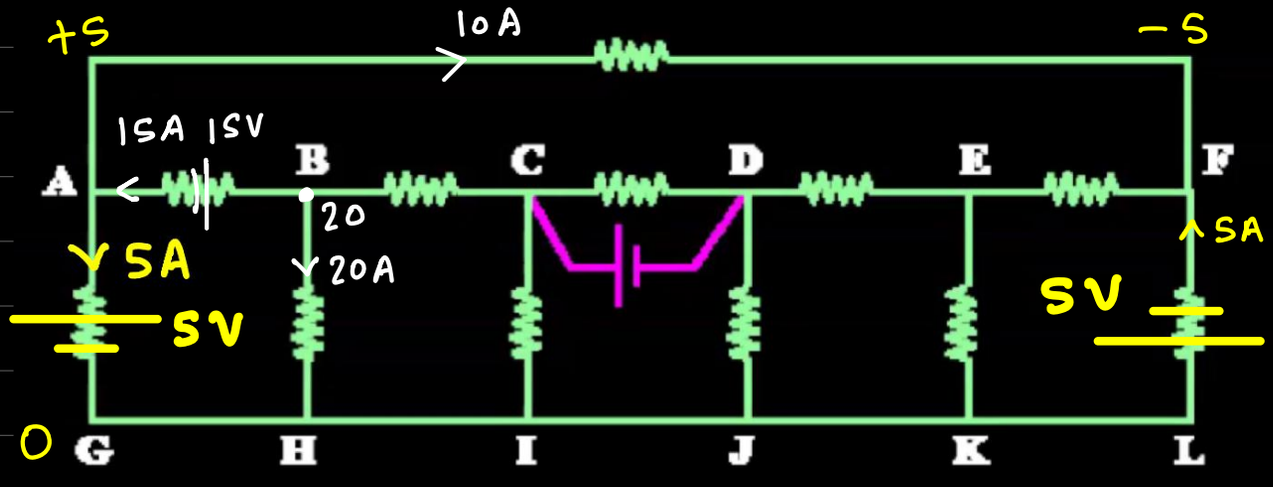
EXAMPLE:

In the figure shown, resistance of each resistor are same. If current in AG is 5A then



(A) current in FL is 5 A
(B) current in BA is 15 A
(C) Current in AF from top resistor is 10 A
(D) current in BH is 20 A

Solution:



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For simplicity we assume all resistors to be 1Ω (Since battery can always be scaled to push a 5A current through AG (1Ω) resistor. By symmetry, EF should also carry 5A. Now by compensation theorem, we replace AG and EF by 5 Volt batteries as shown. (FE is as beautiful to the negative terminal as AG to the positive terminal).

Now current through direct resistor FA = (5-(-5))/1 = 10A

Thus options A and C are correct. Now by junction law at A, current in BA = 15A thus option B is also correct.

Now potential drop in BA = 15A × 1Ω = 15V

Now potential of node B = 5V + 15V = 20V, thus current in BH: (20-0)/1 = 20 Amp. Thus option D is also correct.

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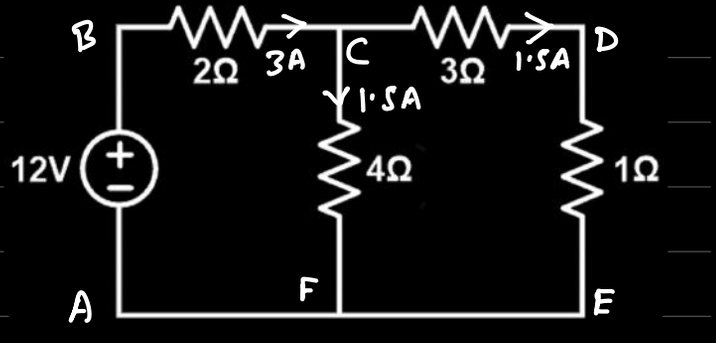
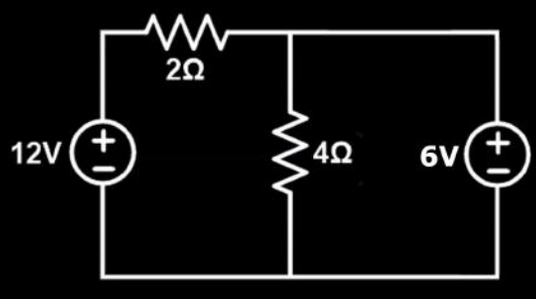
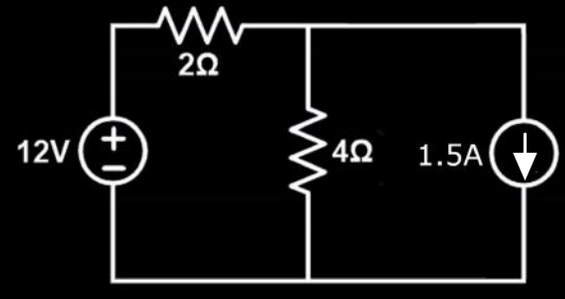
SUBSTITUTION THEOREM: If any element in a network is replaced by a voltage source whose voltage at any instant of time is equal to the voltage across the element in previous network, then the conditions in rest of the network will be unaltered.

Alternately, if an element in a network is replaced by a current source whose current at any instant is equal to the current through the element in the original network then, rest of the network remains unaltered. For instance consider the following circuit:



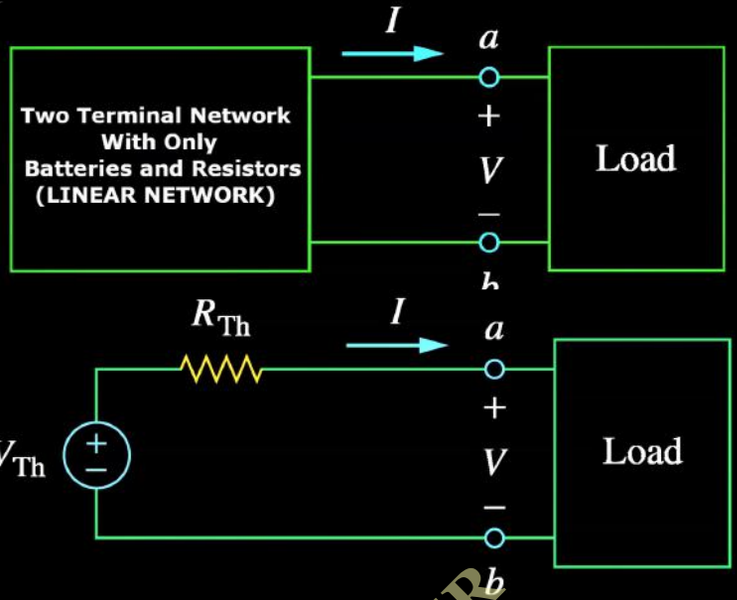
Here total potential drop across branch CDE is 6V and the current is 1·5A, thus it can be replaced either by a current source of 1·5A or a battery of EMF of 6V. Thus following simplifications are allowed:





THEVENIN'S THEOREM:

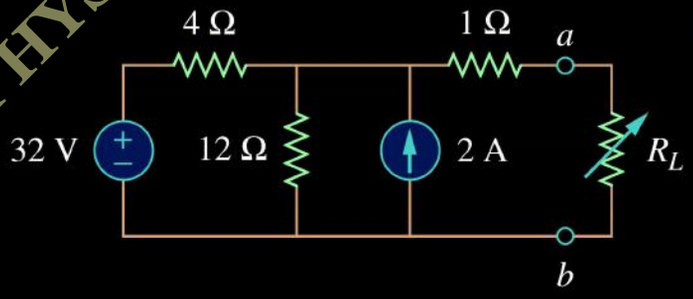
If we have an anbitrary network containing only batteries and resistors connected across a load network through only two ports, then such a network can be replaced by a single ideal Voltage source of EMF V_{TH} and resistance R_{TH} where:



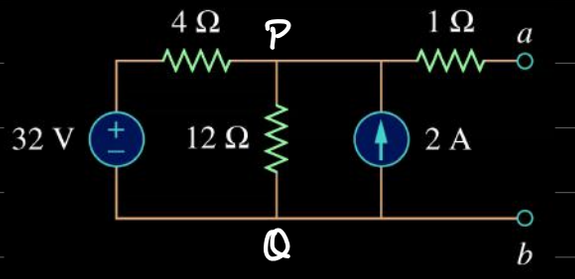
V_{TH}: Open circuit Voltage across the terminals a and b (see figure)

and R_{TH}: Resistance of the linear network across terminals a and b with all batteries replaced by short circuits and current sources replaced by open circuits. This will be clear from following example:

EXAMPLE: In the adjacent circuit, thevenize R_L, hence find current through R_L if $R_L = 6\Omega, 16\Omega$ and 36Ω.



Solution: We will find V_{th} between a and b using superposition. Note that nothing flows through 1Ω resistor in open circuit thus potential difference between a and b in O.C. will be same as that across 12Ω.



Now considering Battery alone (current source considered as open circuit), Drop across PQ: $V_1 = \frac{12}{16} \times 32 = 24\,V$ (P higher).

Now due to current source alone (battery considered short circuit), Current through 12Ω: $I_{12} = \frac{1/12}{1/12 + 1/4} \times 2 = 1/2$ Amp $\Rightarrow$

P.D. across 12Ω, $V_2 = 1/2 \times 12 = 6\,V$ (P higher)

$\Rightarrow V_{th} = V_1 + V_2 = 24 + 6 = 30\,V$ Ans

Now R_{th}: (12||4) Sen (1) C (∵ Current source has infinite resistance). AMBARISH SIR

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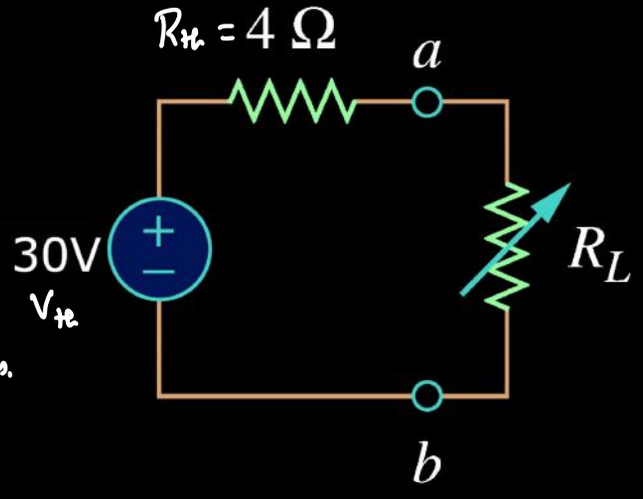
Thus $R_{th} = 4\Omega$. Now equivalent thevenized circuit is shown in the figure. Now when $R_L = 6\Omega$

$i = 30/(6+4) = 3$ A Ans.

For $R_L = 16\Omega$, $i = 30/(16+4) = 1.5$A Ans

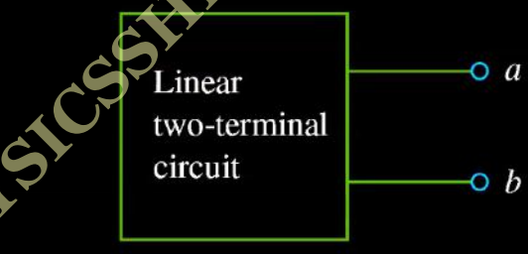
For $R_L = 36\Omega$, $i = 30/(36+4) = 0.75$A Ans.

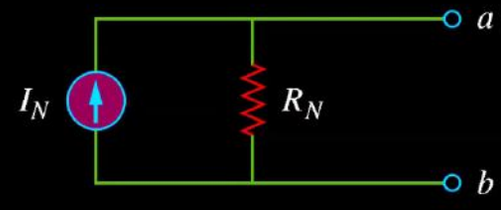


→ Had we not thevenized the R_L, we will have to put considerably more effort solving the circuit from scratch for each value of R_L.

NORTON'S THEOREM: Any linear driven circuit (contains only current sources, batteries and resistors) can be replaced by a single current source and a parallel resistance (see figure) Here the Norton Resistance (R_N) is same as the thevenin resistance and Norton current I_N is the short circuit current between terminals a and b.

($I_N = \mathcal{E}_{th}/R_{th}$)



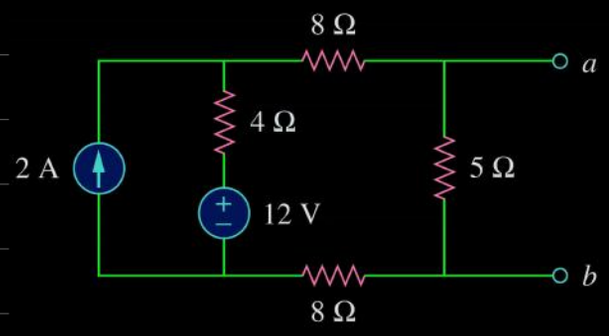


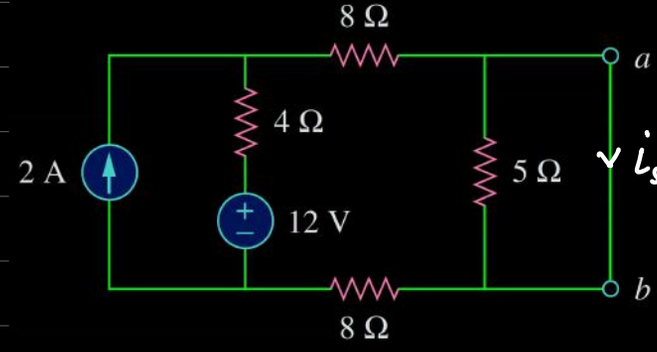
EXAMPLE: Nortonize the shown driven network between terminals a and b.

Solution: We short circuited a and b and now will find current through it by superposition, (Note that 5Ω is redundant for calculating current since it is in parallel to short circuit). Now due to current source alone, current through a-b:

$$i_1 = \frac{1/16}{1/16 + 1/4} \times 2 = 2/5\ A$$

(Battery was treated as S.C.)





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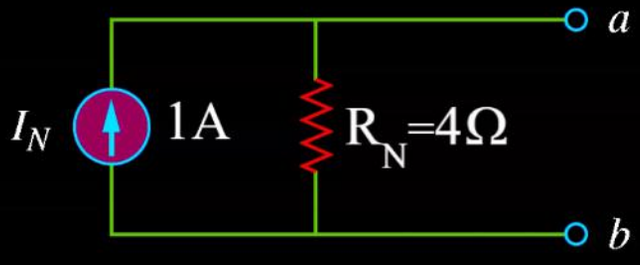
Now due to battery alone, current through a-b: $i_2 = 12/(4+8+8) = 3/5A$

Now total $i_{sc} = i_1 + i_2 = 2/5 + 3/5 = 1A$.

Now $R_N = R_{th} = 5 \,||\, (8+4+8) = 20 \times 5/(20+5) = 4\Omega$

Thus Norton equivalent circuit is as shown in the figure.

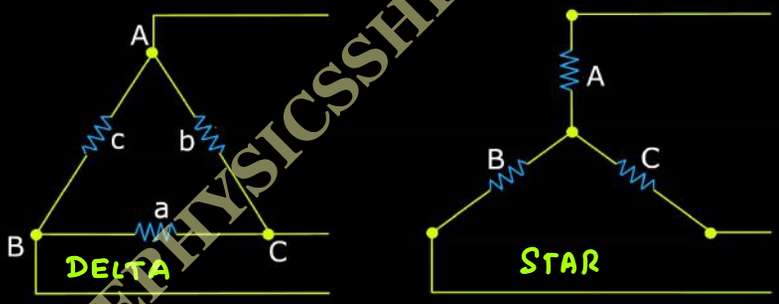


→ Note Thevenin Equivalent is readily inferred from Norton and Vice Versa using the formula, $\mathcal{E}_{th} = I_N R_N \; (= R_{TH})$ thus here $\mathcal{E}_{th} = 4V$.

STAR DELTA TRASFORM:

DELTA TO STAR:

$$A = \frac{bc}{a+b+c} \quad \text{etc.}$$



STAR TO DELTA TRASFORM

$$a = \frac{AB+BC+CA}{A} \quad \text{etc.}$$

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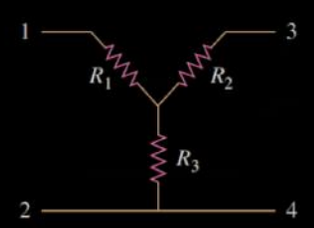
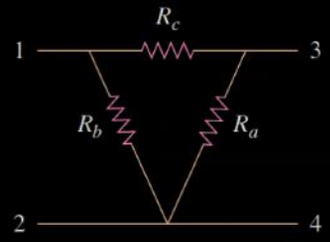
→ For Capacitors also star delta transforms exist and the formula forms swap i.e.

$$a = BC/A+B+C \quad \text{and} \quad A = \frac{ab+bc+ca}{a}$$

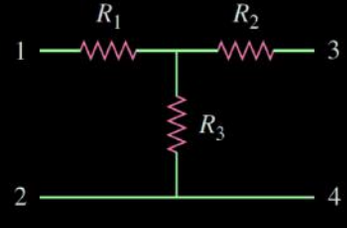
→ Sometimes these transforms are also called Y-Δ or T-π transforms due to alternative looking same topologies:



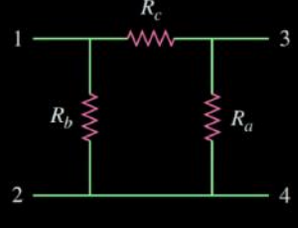
(Y or Star)



(DELTA)



(T or Star)



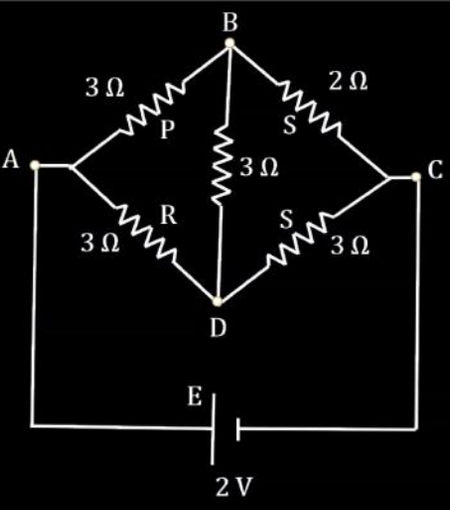
(π or Δ)

EXAMPLE:

For adjacent circuit find effective resistance by:

(a) Starrize the delta ABD

(c) Deltize the node D.

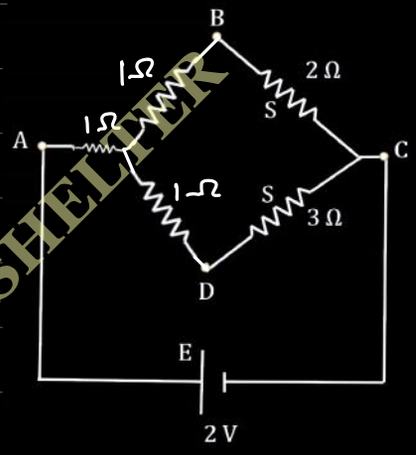


Solution: When we convert Δ ABD to star, each 3Ω resistor is replaced by $\frac{3\times3}{3+3+3}$

i.e 1Ω resistor, thus equivalent star circuit is as shown in the figure.

Thus R_{eff} = (3||4) Ser (1)

$= \frac{12}{7} + 1 = \frac{19}{7}\Omega$ Ans

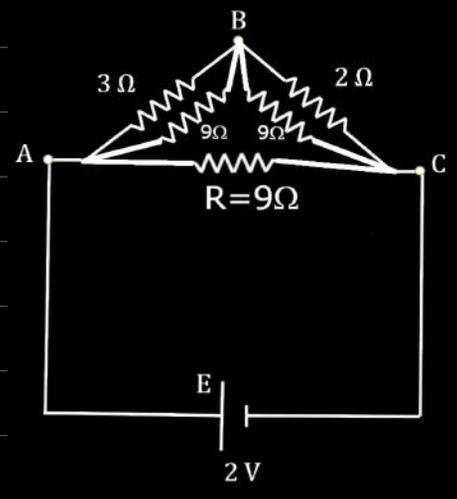


(b) When star at D is converted to Delta, each resistance, $R = (3\times3+3\times3+3\times3)/3 = 9\Omega$

Thus the simplified circuit is shown in the adjacent figure. The effective resistance:

$$[(3||9)\ \text{Ser}\ (2||9)]\ ||\ 9 = \left(\frac{27}{12} + \frac{18}{11}\right)||\ 9$$

$$= \frac{171}{44}\ ||\ 9 = 19/7\ \Omega \quad \text{(Same answer)}$$



HEAT DISSIPATION IN A RESISTOR:

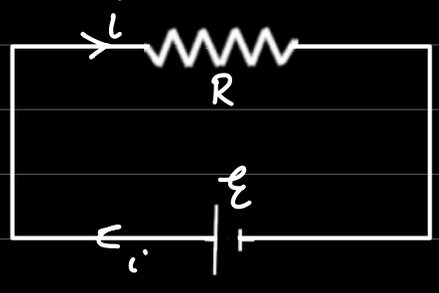
Consider circuit in adjacent figure, writing KVL for the circuit:

$$iR - \mathcal{E} = 0 \Rightarrow \mathcal{E} = iR$$

Now multiplying both sides by i:

$$\mathcal{E}i = i^2R$$, Now:

$\mathcal{E}i = \mathcal{E}\,dq/dt = \frac{dW}{dt}$, thus LHS represents power developed by the battery therefore RHS must represent heat dissipation in the resistor, thus for any resistor,



$$\frac{dH}{dt} = i^2R = Vi = \frac{V^2}{R}$$

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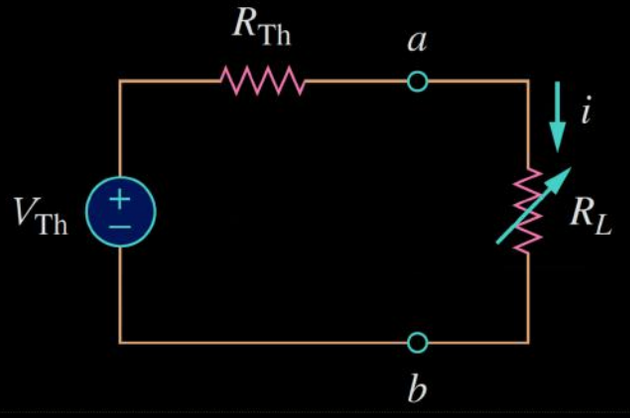
Maximum Power Transfer Theorem:

Let R_{TH} be the thevenin resistance of a driven network that drives current through a variable load resistor R_L. We wish to maximize the power dissipation in R_L. Now:



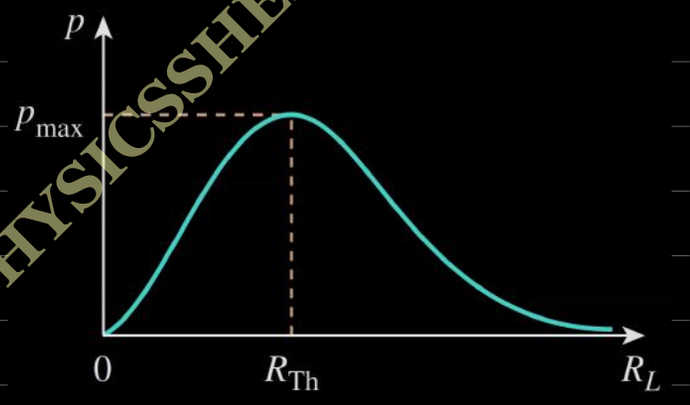
$$i = \frac{V}{R_{TH} + R_L}$$

$\Rightarrow$ Power dissipation in $R_L = i^2 R_L = \dfrac{V_{TH}^2 R_L}{(R_{TH} + R_L)^2} = P$ (Say)

Now for maximum power dissipation, $dP/dR_L = 0$, thus:

$$V_{TH}^2 \frac{(R_{TH}+R_L)^2 - 2R_L(R_{TH}+R_L)}{(R_{TH}+R_L)^4}$$

$$= \frac{V_{TH}^2}{(R_{TH}+R_L)^3}\left[R_{TH} - R_L\right] = 0$$



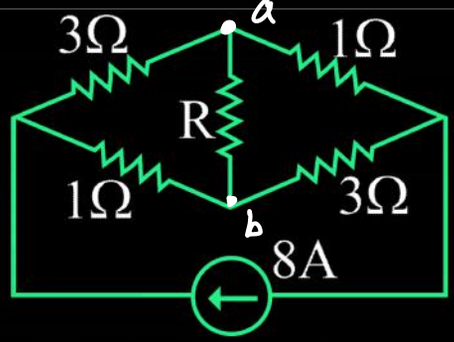
$\Rightarrow \boxed{R_L = R_{TH}}$ Thus for maximum power dissipation, load resistance must be equal to thevenin resistance of the driven circuit.

→ At this condition: $dP/dR_L = 0$, Power is nearly independent of small fluctuations in R_L.

→ Note that this is not the condition for maximum efficiency (in fact for maximum efficiency, $R_L \to \infty$ (obviously, think!)

→ For a given R_L if we had to choose R_{TH} for maximum dissipation in R_L then $R_{TH} \neq R_L$, infact then R_{TH} must be chosen to be zero).

Example: For the circuit in adjacent figure, find the value of R for maximum power dissipation in it.



Solution: We need to find R_{TH} across R. (Remember that a current circuit is replaced by an open circuit). Now after removing R across

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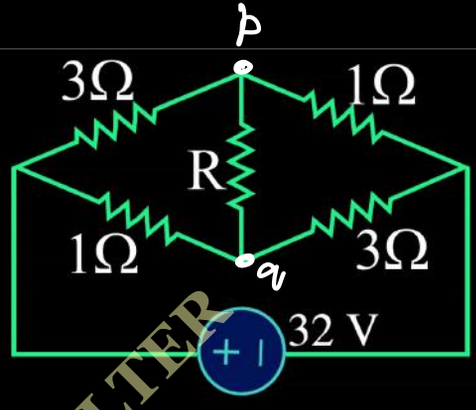
a and b and replacing current source by open circuit, $R_{ab} = R_{th} = (3 \text{ ser } 1) \,||\, (1 \text{ ser } 3) = 2\Omega$, thus required value of

$R = R_H = 2\Omega$ Ans.

EXAMPLE: For shown adjacent circuit find the value of R for maximum power dissipation in it.

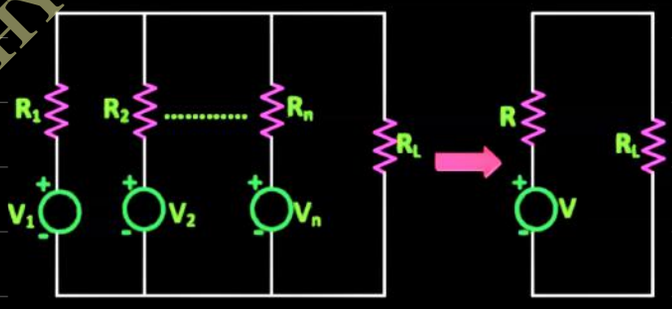


Solution: For finding R_{TH} we replace battery by short circuit. Thus nodes A and B can be collapsed to a single point, also R is removed for calculating R_{TH}.

Now $R_{TH} = R_{PQ} = (3\,||\,1) \text{ ser } (1\,||\,3) = 1.5\Omega$, thus require value,

$R = R_{TH} = 1.5\Omega$ Ans.

MILLMAN'S THEOREM: If we have a network of several voltage sources connected in parallel driving some current across a load resistor, then Entire network can be reduced to a single battery and single resistor in series with that battery as shown, there:



$$V = \frac{V_1/R_1 + V_2/R_2 + V_3/R_3 + \cdots}{1/R_1 + 1/R_2 + 1/R_3 + \cdots}$$

and $$\frac{1}{R} = \frac{1}{R_1} + \frac{1}{R_2} + \frac{1}{R_3} + \cdots$$

→ Note that V_i are signed values, if upward polarity is considered as positive then downward polarity battery will be considered as negative.

→ Millman's theorem can be generalized to mixed parallel branches containing current and voltage sources simultaneously also, the formula for mixed sources becomes:

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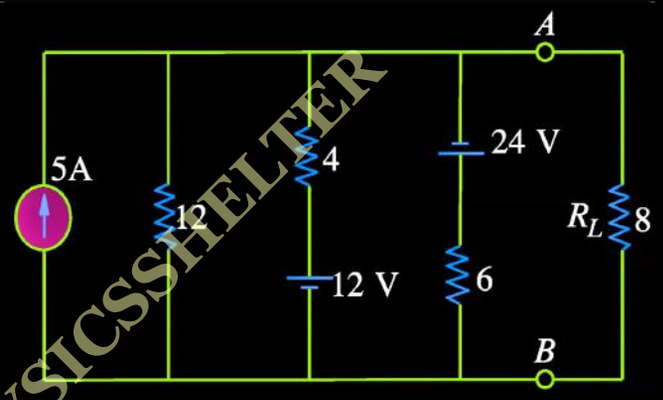
$$V = \frac{\sum V_i/R_i + \sum I_j}{\sum 1/R_i} \quad \text{and} \quad R_{eff} = \left(\sum 1/R_i\right)^{-1}$$

→ This can be directly inferred from the fact that current source has infinite resistance and infinite EMF such that $\mathcal{E}/R = I_j$ for j^{th} current source. (V_i is the voltage of i^{th} battery and R_i its series resistance). For current source, a series resistance will not matter because it anyway has infinite resistance.

EXAMPLE:

For circuit in adjacent figure calculate Millman Voltage and Resistance across R_L. Hence find the current through R_L.



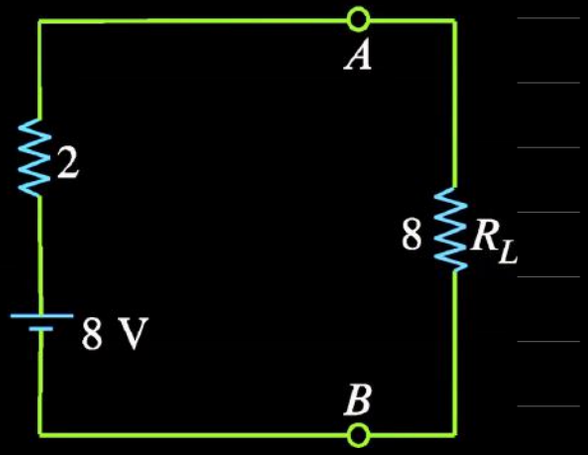
Solution:

$$V = \frac{5 + 0/12 + 12/4 - 24/6}{1/12 + 1/4 + 1/6} = \frac{60 + 36 - 48}{1 + 3 + 2} = 8\text{V}$$

$$R = \left(\frac{1}{12} + \frac{1}{4} + \frac{1}{6}\right)^{-1} = 2\,\Omega$$

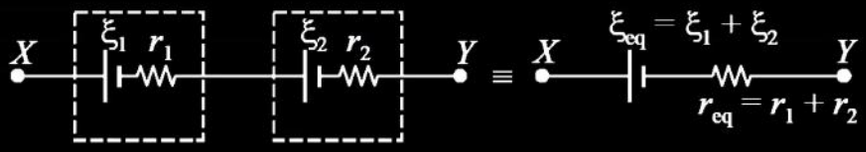
Thus effective millman simplification is redrawn in adjacent figure. Thus required current through load resistance:

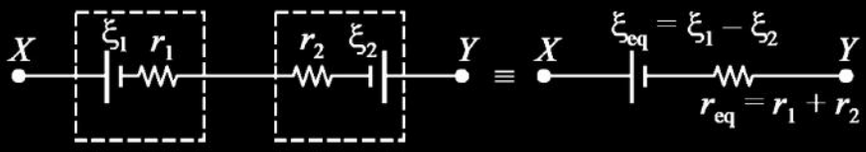
$$i = 8/(6+2) = 0.8\text{ A}$$ Ans.



SERIES GROUPING OF CELLS:

In series combination of cells, the EMFs are simply added algebraically (keep in mind whether batteries are supporting or opposing). Effective resistance is simply added. (For example adjacent diagram shows combination of two cells, same idea can be extended to any





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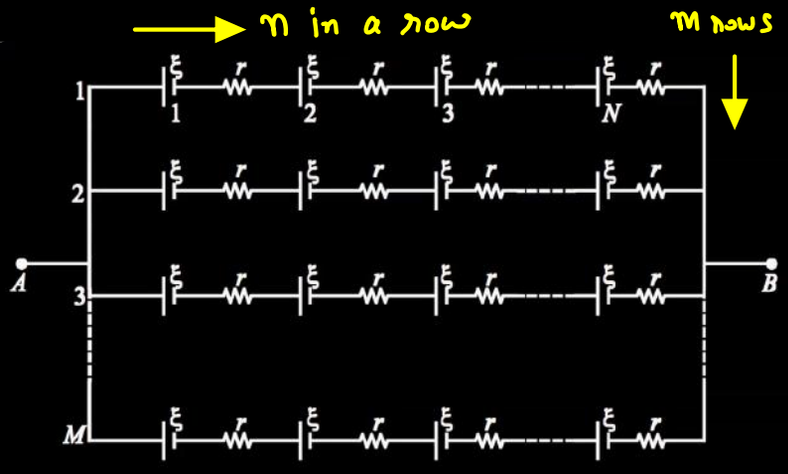
number of cells).



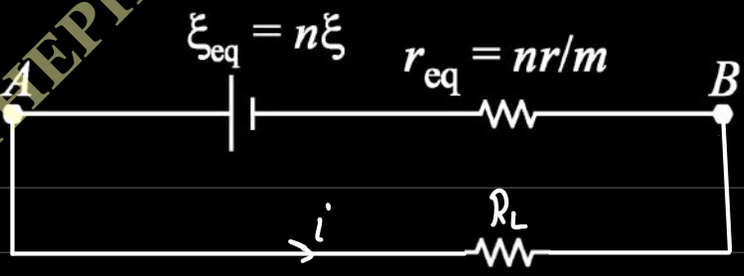
BATTERY GRID: From our discussion on Millman's theorem and series groupings of cells we can readily find the equivalent simplification of grid as:

$$\mathcal{E}_{eff} = \sum_{j=1}^{m} \frac{n\mathcal{E}}{nr} \Big/ \sum_{j=1}^{m} \frac{1}{nr} = n\mathcal{E}$$

and $r_{eff} = nr/m$.

MOST EFFECTIVE GRID FOR MAXIMUM CURRENT IN R_L:

Suppose we have a total of N cells which we wish to arrange in to an m×n grid (n in a row) so as to drive maximum current through R_L



Then: $$i = \frac{n\mathcal{E}}{nr/m + R_L} = \frac{mn\mathcal{E}}{nr + mR_L} = \frac{N\mathcal{E}}{\left(\sqrt{nr} - \sqrt{mR_L}\right)^2 + 2\sqrt{Nr\,R_L}}$$

($\because mn = N$), now this is maximum if $nr = mR_L$, thus for maximum current in the circuit:

$$\boxed{nr = mR_L} \quad (1)$$

→ Now $mn = N$ and equation (1) can be used to solve for m and n (provided we get integer values of m and n).

→ In this manner, most effective combination requires, cells be arranged in such a manner that their effective resistance becomes equal to external resistance (cells are identical)

→ Caution! If we have to choose R_L for maximum current through it, we should have $R_L = 0$.

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EXAMPLE: You have 36 cells of internal resistance 0.5Ω and EMF 1.5 each. How should you arrange them to get maximum current through a 2Ω resistor, also find this maximum current.

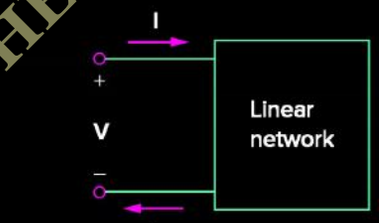
Solution: $\frac{nR}{m} = R_L \Rightarrow \frac{n}{(36/n)} \times 0.5 = 2 \Rightarrow n^2 = 36 \times 4 \Rightarrow n = 12$

Thus we need to create a 12×3 grid (12 in a row and 3 rows).

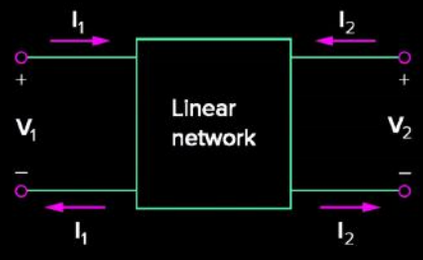
Now max current: $\frac{n\mathcal{E}}{2R_L}$ $(\because R_L = nR/m) \Rightarrow i_{max} = \frac{12 \times 1.5}{2 \times 2}$

$\Rightarrow i_{max} = 4.5\ A$ Ans

TWO PORT NETWORK: A port is just a pair of terminals through which a current may enter or leave. We have mostly dealt with one port networks so far, however we can also have a two port network, i.e two pairs of terminals for entry and exit of current. Charge conservation requires current entering a port must be equal to current leaving that port. Further a network is said to be linear if it contains only batteries and resistors (most common) [Though linearity can also be satisfied by a complex LCR circuit also in context of alternating current circuits. (L = Inductance). In a two port network, let input side current and voltage be I_1 and V_1 respectively and let corresponding values for output port be I_2 and V_2 then due to linearity, out of the four variables I_1, I_2, V_1 and V_2 any two can be expressed as linear combinations of other two quantities with coefficients being constants depending on the network inside the box from which four ports emerge.



ONE PORT NETWORK



TWO PORT NETWORK

IMPEDANCE PARAMETERS (RESISTANCE PARAMETERS): When the voltages V_1 and V_2 are expressed as linear combination of currents I_1 and I_2 the coefficients has dimensions of resistance and these coefficients

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are called Impedance ("Resistance") parameters. We can write:

$$V_1 = Z_{11} I_1 + Z_{12} I_2$$

and $$V_2 = Z_{21} I_1 + Z_{22} I_2$$ or in matrix notation:

$$\begin{bmatrix} V_1 \\ V_2 \end{bmatrix} = \begin{bmatrix} Z_{11} & Z_{12} \\ Z_{21} & Z_{22} \end{bmatrix} \begin{bmatrix} I_1 \\ I_2 \end{bmatrix}$$ or in short,

$[V] = [Z][I]$ where $[V]$, $[Z]$ and $[I]$ represent the corresponding matrices. We can also work out the inverse relation using:

$[I] = [Z^{-1}][V]$ where Z^{-1} is the inverted Z matrix (Also known as admittance (Y) matrix).

$Z_{11} = \left.\frac{V_1}{I_1}\right|_{I_2=0}$ i.e open circuit input resistance

$Z_{12} = \left.\frac{V_1}{I_2}\right|_{I_1=0}$ i.e. open circuit transfer impedance from port 1 to 2

$Z_{21} = \left.\frac{V_2}{I_1}\right|_{I_2=0}$ i.e. open circuit transfer impedance from port 2 to 1.

$Z_{22} = \left.\frac{V_2}{I_2}\right|_{I_1=0}$ i.e open circuit output resistance (at port 2).

A detailed theory of two port networks is covered in electrical engineering courses, here we are just stating the results without bothering to prove (proof are essentially based on superposition theorem).

→ If the two port network does not contain any battery / current sources, we call such networks reciprocal and for such networks $Z_{12} = Z_{21}$. Also for such networks excitation and response follow reciprocal relationship.

→ As a simple example consider the two port networks on the next page. Here in the first network battery is connected on left port and ammeter on the right side, while in the second network the positions of battery and ammeter are swapped

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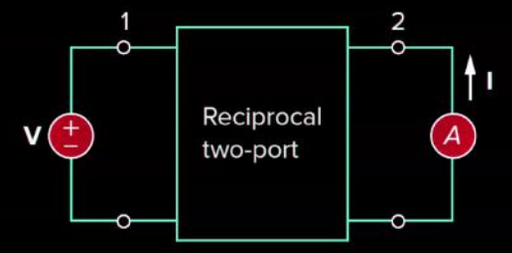
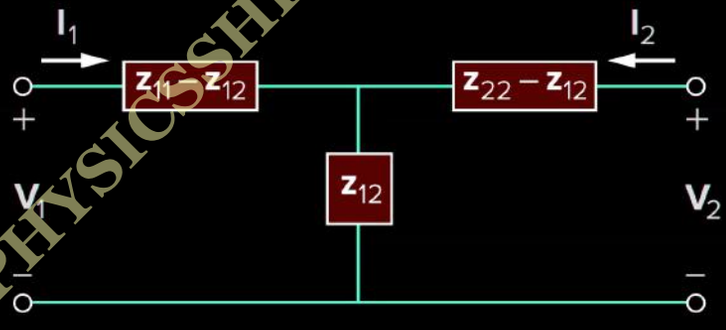
however if the box contains only resistors, the reading of ammeter will remain the same.

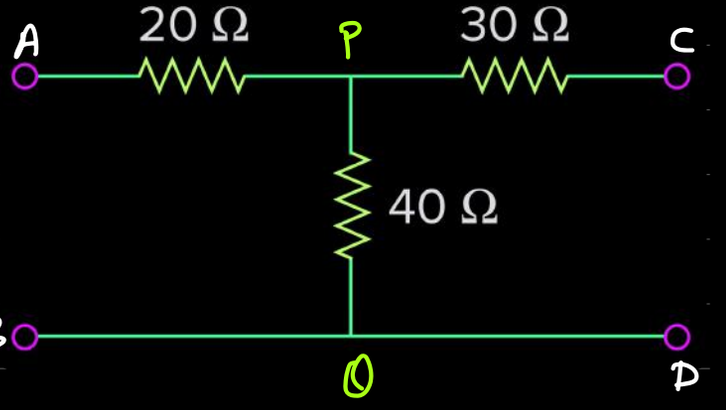


RECIPROCITY THEOREM: If in any circuit containing only one battery, battery is swapped with the position of ideal ammeter, then the reading of ammeter remains unchanged. [Similarly if positions of a single current source and ideal voltmeter are swapped the readings remain unchanged].

T-EQUIVALENT NETWORK: Any complicated two port network can always be replaced by a network of three resistors (provided no battery in the box). The adjacent figure shows the required resistance values for the two port network equivalent.



EXAMPLE: Find the impedance parameters for the adjacent network using basic definitions.



Solution: $Z_{11} = \left.\frac{V_1}{I_1}\right|_{I_2=0}$

i.e input resistance with output terminals open. Thus $Z_{11} = 20 + 40 = 60\,\Omega$

Similarly $Z_{22} = \left.\frac{V_2}{I_2}\right|_{I_1=0}$; i.e $30 + 40 = 70\,\Omega$

Now $Z_{12} = \left.\frac{V_1}{I_2}\right|_{I_1=0}$, Imagine $I_2 = 1\,A$ and a voltmeter across A-B, Thus Z_{12} = Voltage across 40 V / 1A thus,

$Z_{12} = 40\,\Omega$

Also $Z_{21} = \frac{V_2}{I_1} = Z_{12}$ (∵ no battery) thus $Z_{21} = 40\,\Omega$

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Thus $Z = \begin{bmatrix} 60 & 40 \\ 40 & 70 \end{bmatrix}$

EXAMPLE : KROTOV : 3.34

A concealed circuit (black box) consisting of resistors has four terminals. If a voltage is applied between clamps 1 and 2 when clamps 3 and 4 are disconnected, the power liberated is $P_1 = 40$ W, and when the clamps 3 and 4 are connected, the power liberated is $P_2 = 80$ W. If the same source is connected to the clamps 3 and 4, the power liberated in the circuit when the clamps 1 and 2 are disconnected is $P_3 = 20$ W. Determine the power P_4 consumed in the circuit when the clamps 1 and 2 are connected and the same voltage is applied between the clamps 3 and 4.

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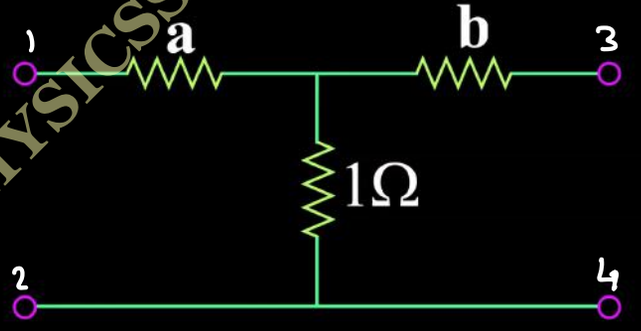
Solution: Recall that power in any network is $P = V^2/R_{Eff}$. For our network we propose the adjacent T network. We have considered one resistance as 1Ω because we can always scale up the voltage to get the desired powers.



(The ratio of powers will only depend on ratios of effective resistances while actual value will be proportional to square of voltage).

Now let our voltage be V. Then, when V is between ① & ② with 3 and 4 open circuit:

$$P_1 = 40\,W = V^2/(a+1) \quad ①$$

With V between ① & ② and 3, 4 shorted,

$$P_2 = 80\,W = \frac{V^2}{a + \frac{b}{b+1}} = \frac{V^2(b+1)}{a+b+ab} \quad ②$$

Now with V between 3 and 4 and 1 & 2 open circuit,

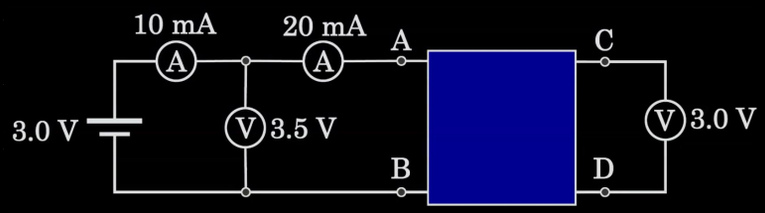
$$P_3 = 20\,W = \frac{V^2}{(b+1)} \quad ③$$

Solving ①, ② and ③: $V = 2\sqrt{10}$, $a = 0$, $b = 1$, Now when V is between ③ & ④ and 1 & 2 shorted, $P_4 = \frac{V^2}{b + a/a+1} = \frac{V^2(a+1)}{a+b+ab}$

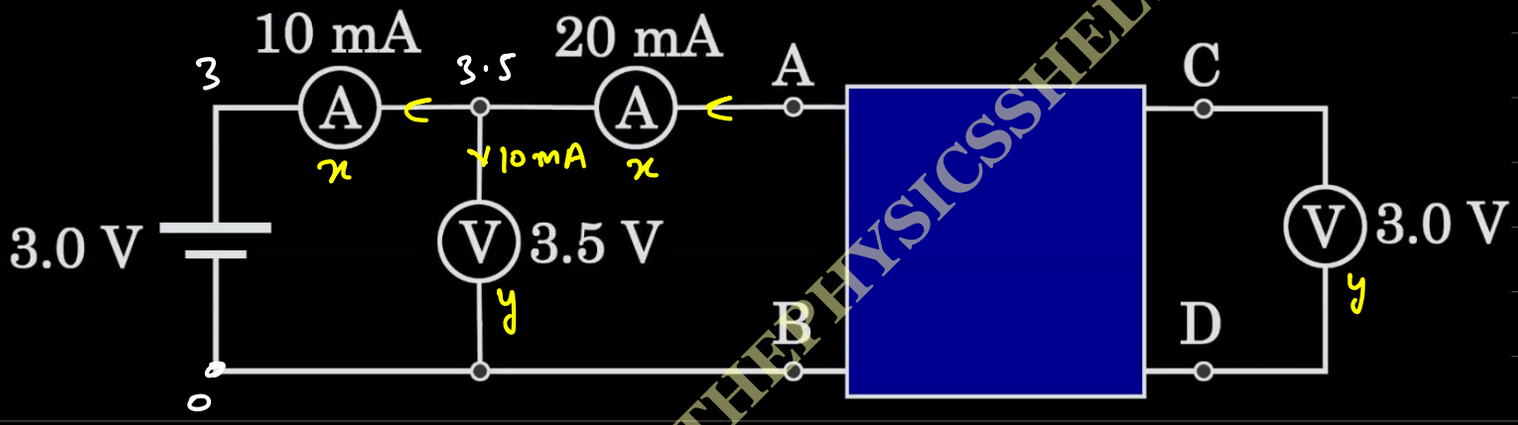
$\Rightarrow P_4 = 40\,W$ Ans.

EXAMPLE: BUILD-17 (PATHFINDER)

A black-box (you cannot see what is inside the box) with an unknown electric circuit inside has four terminals A, B, C and D. To know the contents inside the black-box, following circuit consisting of an ideal battery, two non-ideal identical voltmeters and two non-ideal identical ammeters is arranged. Terminal voltage of the battery and readings of the instruments are shown in the figure. Suggest the simplest possible circuit in the black-box and find its parameters.



Solution:



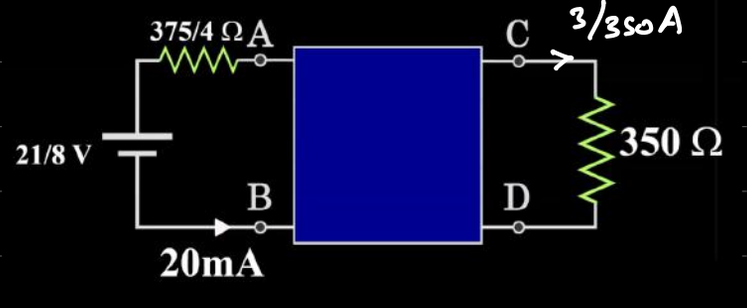
Solution: Let the resistance of ammeters be x and that of voltmeters be y. Now across left ammeter, $i = 10\,mA$, $V = 0.5$ Volt $\Rightarrow x = \frac{0.5}{10^{-2}}$

$\Rightarrow x = 50\,\Omega$, similarly $y = \frac{3.5}{10^{-2}} = 350\,\Omega$.

Now we can thevenize the network to the left of AD for easy analysis. $\mathcal{E}_{th} = \frac{350}{350+50} \times 3 = 21/8$ Volt.

$R_{th} = [xy/(x+y)] + x = 375/4\,\Omega$.

Now the simplified circuit looks as shown in adjacent figure. Our black box needs to satisfy two conditions, i.e. input current 20 mA and output current of 3/350 Ampere (∵ Voltage was 3 Volts there). Thus we can manage with two unknown elements in the circuit. We propose the following circuit:



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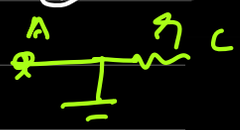
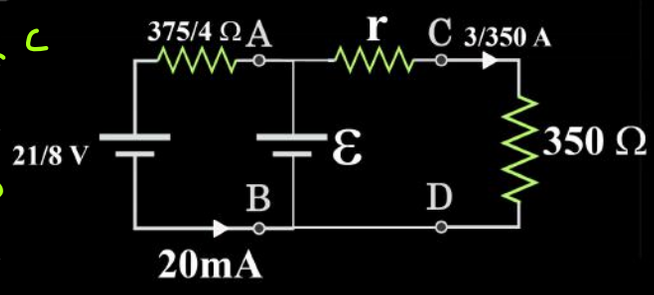
Thus: $\mathcal{E}/(r+350) = 3/350$ ①

also: $\dfrac{\mathcal{E} - 21/8}{375/4} = 20\times10^{-3}$ ②





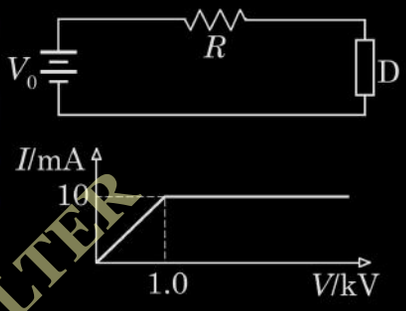
Solving we get: $\mathcal{E} = 4.5\ V$ Ans.

$r = 175\ \Omega$ Ans.

EXAMPLE: PATHFINDER MCQ 6:

Volt-ampere (V-I) characteristics of an unknown device D connected in a circuit in series with a resistance and a battery is shown in the figure. If value of the resistance is $R = 300\ \text{k}\Omega$ and terminal voltage of the battery is $V_0 = 5\ \text{kV}$, find potential drop across the device.

(a) 1.0 kV (b) 2.0 kV
(c) 3.0 kV (d) 4.0 kV

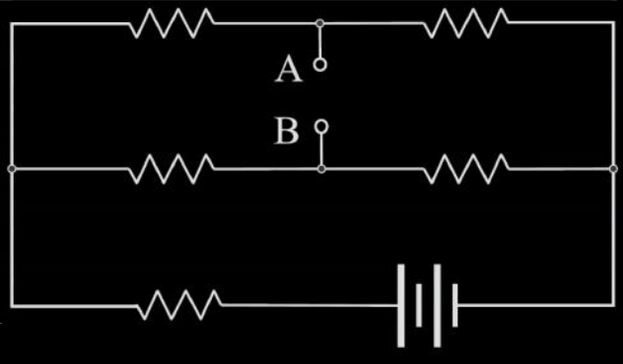


Solution: There are two possibilities: either current is 10 mA or lesser and we will eliminate one of them by logic. Assuming a current of 10 mA drop across 300 kΩ becomes 3 kV, thus for an applied voltage of 5 kV, drop across D must be (5-3) kV i.e 2 kV which is sufficient to cause 10 mA current, so this is definitely possible. However if current is less than 10 mA, Drop across D is less than 1 kV and that across R less than 3 kV and they will not add up to 5 kV, thus only possibility is $V_D = 2\ \text{kV}$ (b) Ans.

EXAMPLE: PATHFINDER MCQ 8

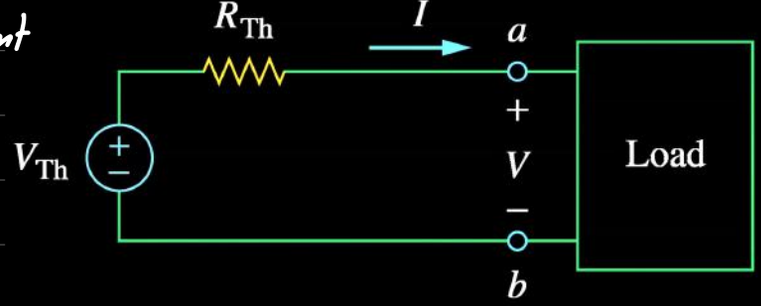
A network shown in the figure consists of a battery and five unknown resistors. When an ideal ammeter is connected between the terminals A and B, its reading is 4 A and when a resistance of 3 Ω is connected in series with the ammeter its reading becomes 2 A. Now the ammeter and the resistance are disconnected and an ideal voltmeter is connected between the terminals A and B. What would the voltmeter read?

(a) 6 V (b) 10 V
(c) 12 V (d) 18 V



Solution: Imagine a thevenin equivalent between A and B then, according to Question: $\dfrac{V_{TH}}{R_{TH}} = 4$ ①

Also $\dfrac{V_{TH}}{R_{TH}+3} = 2$ ②



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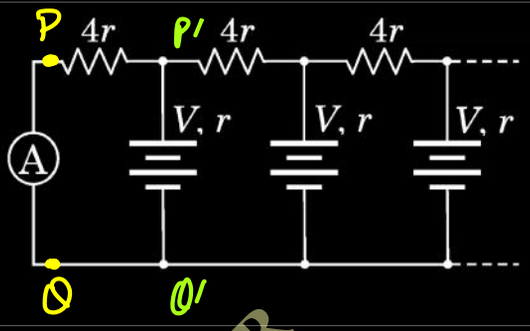
From ① & ②, $V = 12\,V$, $R_{34} = 3\,\Omega$, Now when ideal voltmeter (∞ resistance) is connected, it will just read thevenin voltage i.e 12 V. Thus correction option is (c) Ans.

EXAMPLE : PATHFINDER MCQ - 11

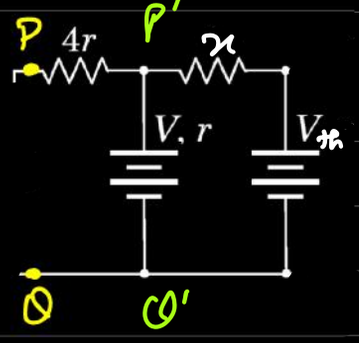
The circuit shown in the diagram extends to the right into infinity. Each battery has electromotive force V (unknown) and internal resistance r (known). Each resistor has resistance $4r$. Reading of the ideal ammeter shown in the diagram is I. Find the value of V in terms of I and r.

(a) $(2-2\sqrt{2})\,Ir$ (b) $(2+2\sqrt{2})\,Ir$
(c) $4.5Ir$ (d) $3.0Ir$



Solution: Thevenizing the ammeter, R_{th} = Resistance of infinite ladder with batteries shorted. Let thevenin resistance be x, then using infinite ladder theory (discussed in capacitors), $\frac{xr}{x+r} + 4r = x$

$\Rightarrow x = 2(\sqrt{2}+1)r$, Now let V_{th} be the effective thevenin voltage of infinite ladder then leaving one unit, still the remaining ladder with have thevenin voltage V_{th}. Now using Millman,

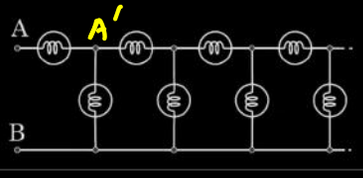


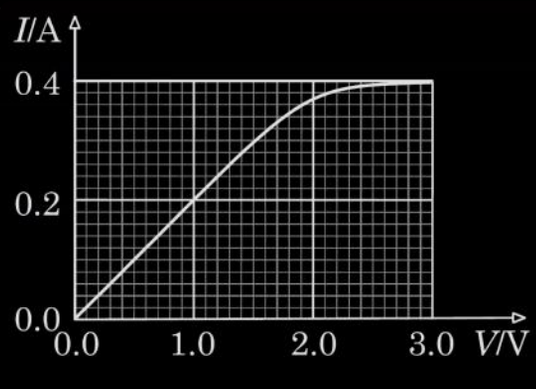
$$\frac{\frac{V_{th}}{x} + \frac{V}{r}}{\frac{1}{r} + \frac{1}{x}} = V_{th} \Rightarrow V_{th} = V$$

Now $I = \frac{V_{th}}{x} = \frac{V}{x} \Rightarrow V = Ix = 2I(\sqrt{2}+1)r$ (b) Ans.

EXAMPLE : PATHFINDER : CHECK - 9:

Volt-ampere (V-I) characteristics of an incandescent lamp is shown in the adjoining figure. This lamp can withstand a maximum voltage 3.0 V and a maximum current 0.4 A. An infinite ladder network consisting of a large number of these lamps is also shown in the figure given below. How much maximum voltage can be applied between the terminals A and B, so that no lamp gets damaged? (Use $\sqrt{5} \approx 2.2$).





Solution: We observe that system has a nearly ohmic behavior till about 1.5 Volts and after that non ohmic behaviour.

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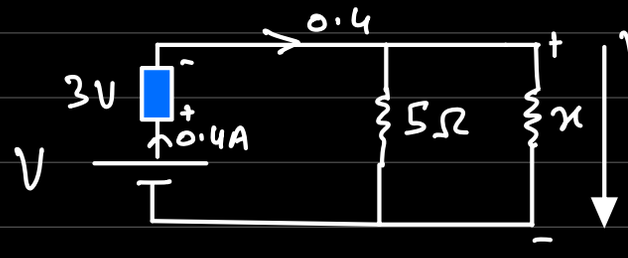
In the ladder maximum current will flow through the bulb between A and A' and then the current will branch out and other bulbs will have smaller currents. In the limiting case, the bulb between A-A' will have a current of 0.4A and potential difference of 3V. In the ohmic domain, the resistance of each bulb is 5Ω. Now let the resistance of infinite ladder after A' be x, then by standard theory of infinite ladder:

$$\frac{5x}{5+x} + 5 = x \Rightarrow x = \frac{5(\sqrt{5}+1)}{2}$$

Now let required maximum voltage be V, applying compensation for bulb across A-A', our circuit looks like:



Now drop across 5Ω should be less than 3V.

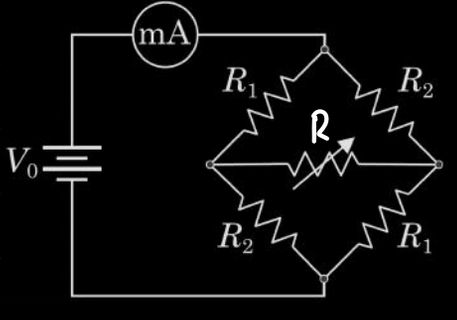
$\Rightarrow$ Now Drop across // combination of 5Ω & x:

$$V_0 = \left(\frac{5x}{5+x}\right) \times 0.4 = 1.2 \text{ Volt}$$

Thus $V = 3 + 1.2 = 4.2$ Volts Ans.

EXAMPLE : PATHFINDER : CHECK 4 :

A network shown in the figure consists of an ideal battery of terminal voltage $V_0 = 3.00$ V, a milli-ammeter of very low resistance, four fixed value resistances and a rheostat. If on varying resistance of the rheostat from a large value to a vanishingly small value, reading of the milli-ammeter varies between $I_1 = 0.75$ mA to $I_2 = 1.00$ mA, find values of the fixed resistances R_1 and R_2.



Solution: When R is very large, $R_{eff} = \frac{1}{2}(R_1 + R_2)$

$\Rightarrow \frac{3}{\frac{1}{2}(R_1+R_2)} = 0.75$ mA (1) now when $R \to 0$, $R_{eff} = \left(\frac{R_1 R_2}{R_1+R_2}\right) \times 2$

Thus $\frac{3}{2\left(\frac{R_1R_2}{R_1+R_2}\right)} = 1$mA (2) Solving (1) & (2), $R_1 = 2k\Omega$, $R_2 = 6k\Omega$ or $R_2 = 2k\Omega$ & $R_1 = 6k\Omega$ Ans.

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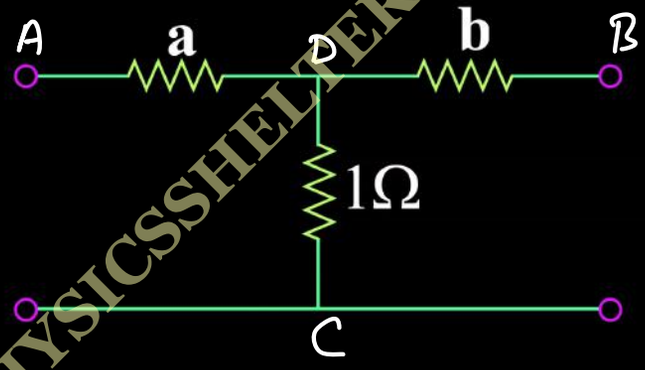
Example : Pathfinder : Check-2 :

A black-box has three terminals A, B and C coming out. If across A and B a voltage of 20 V is applied, the terminals B and C show 8 V across them. If across the terminals B and C a voltage of 20 V is applied, the terminals A and C show 15 V across them. If the black-box contains only resistances, suggest simplest circuit inside the black-box.

Solution: We have two conditions so our network can have two unknowns so that we may be able to solve for them uniquely. So let us propose a T network with two unknown resistances. Note that voltmeter acts like an open circuit. So when $V_{AB} = 20\ V$,



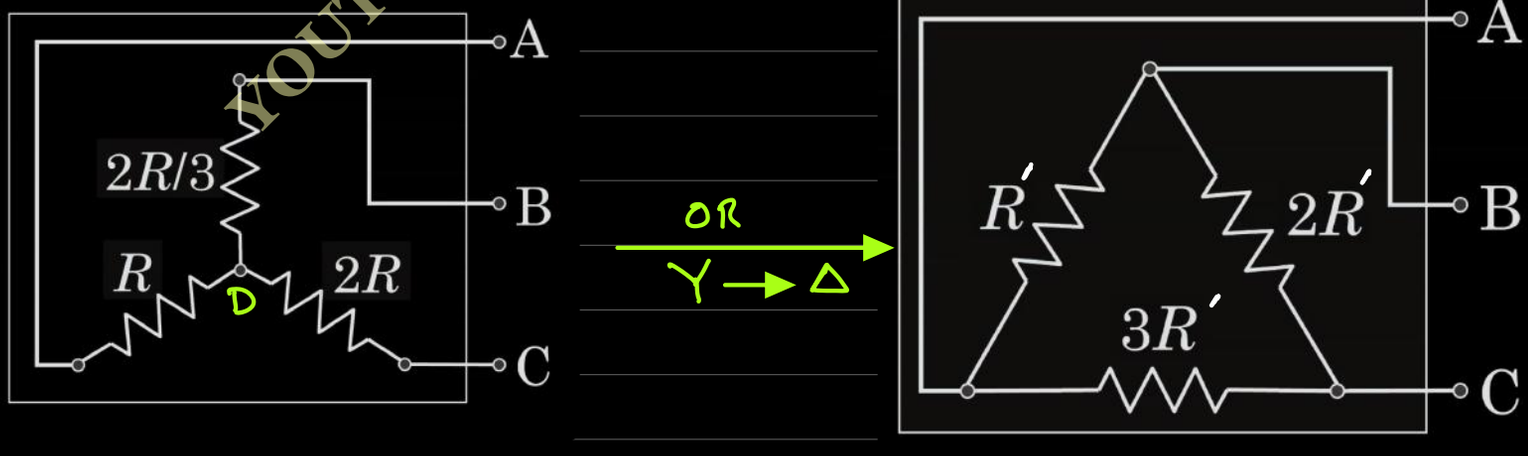
$$V_{BC} = V_{BD} = \frac{b}{a+b} \times 20 = 8$$

$$\Rightarrow \frac{b}{a+b} = \frac{2}{5} \quad (1)$$

Now when $V_{BC} = 20\ V$, $V_{AC} = V_{DC}$

$$= \frac{1}{1+b} \times 20 = 15 \quad (2)$$

Now Solving (1) & (2): $\Rightarrow a = 1/2$, $b = 1/3$. Now the voltages depend only on ratio of resistances rather than their absolute values, we only need to ensure $R_{AD} : R_{DC} : R_{DB} = \frac{1}{2} : 1 : 1/3$ or $R : 2R : \frac{2}{3}R$. Thus equivalent network can be made as:



Where $R' = 2R$.

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